

BFS Traversal

Step 1: Start at E

Legend



Complete



Processing



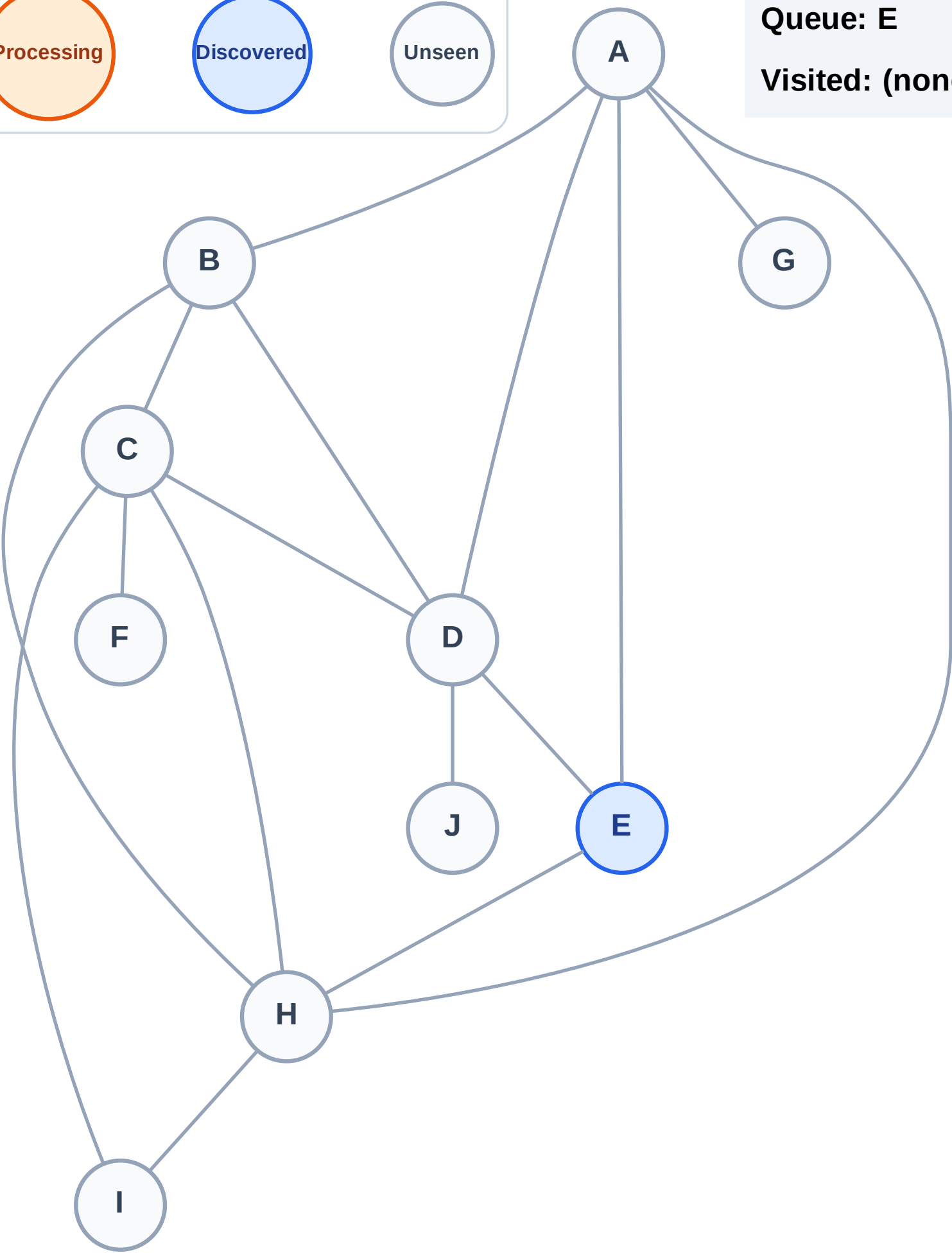
Discovered



Unseen

Queue: E

Visited: (none)



BFS Traversal

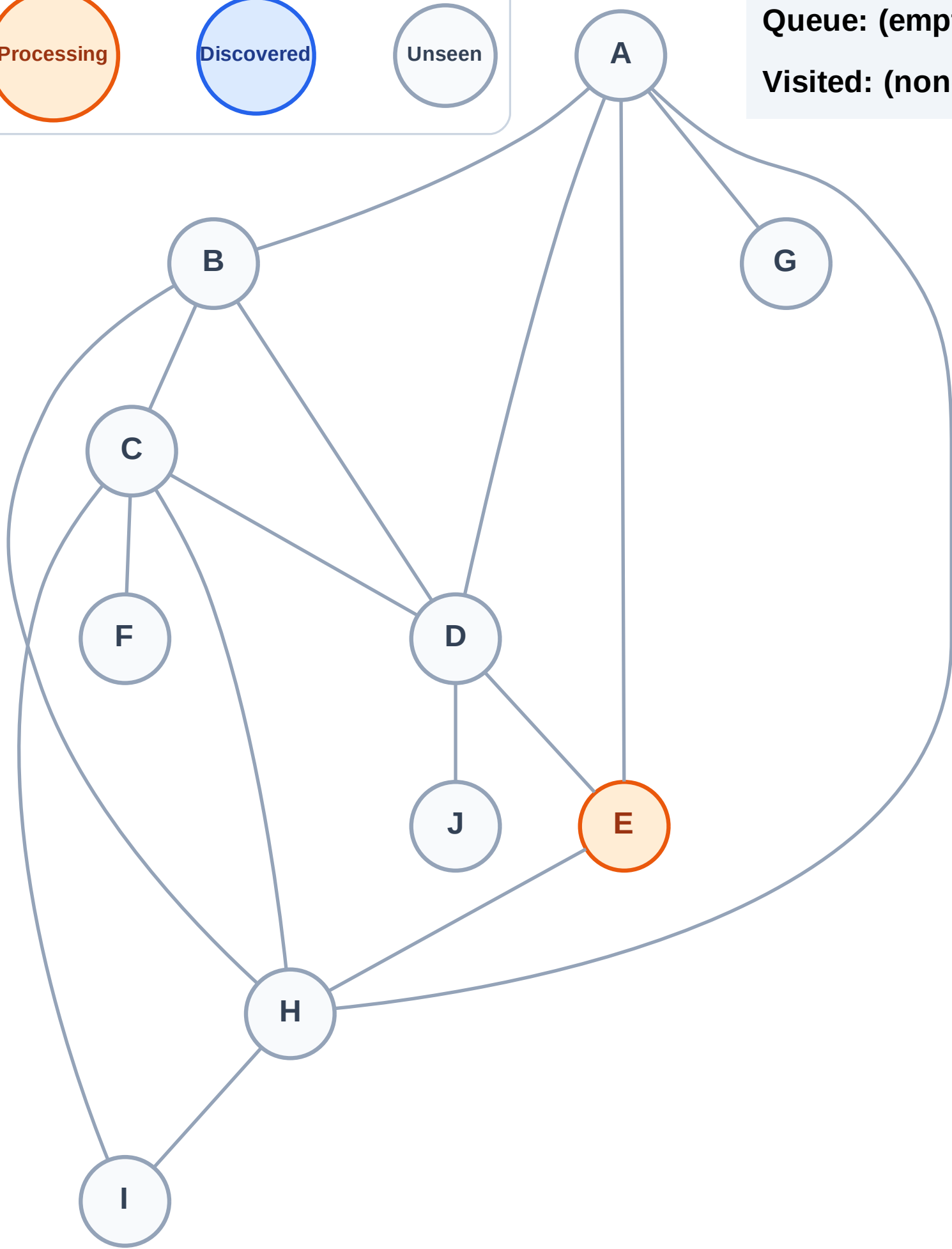
Step 2: Process node E

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

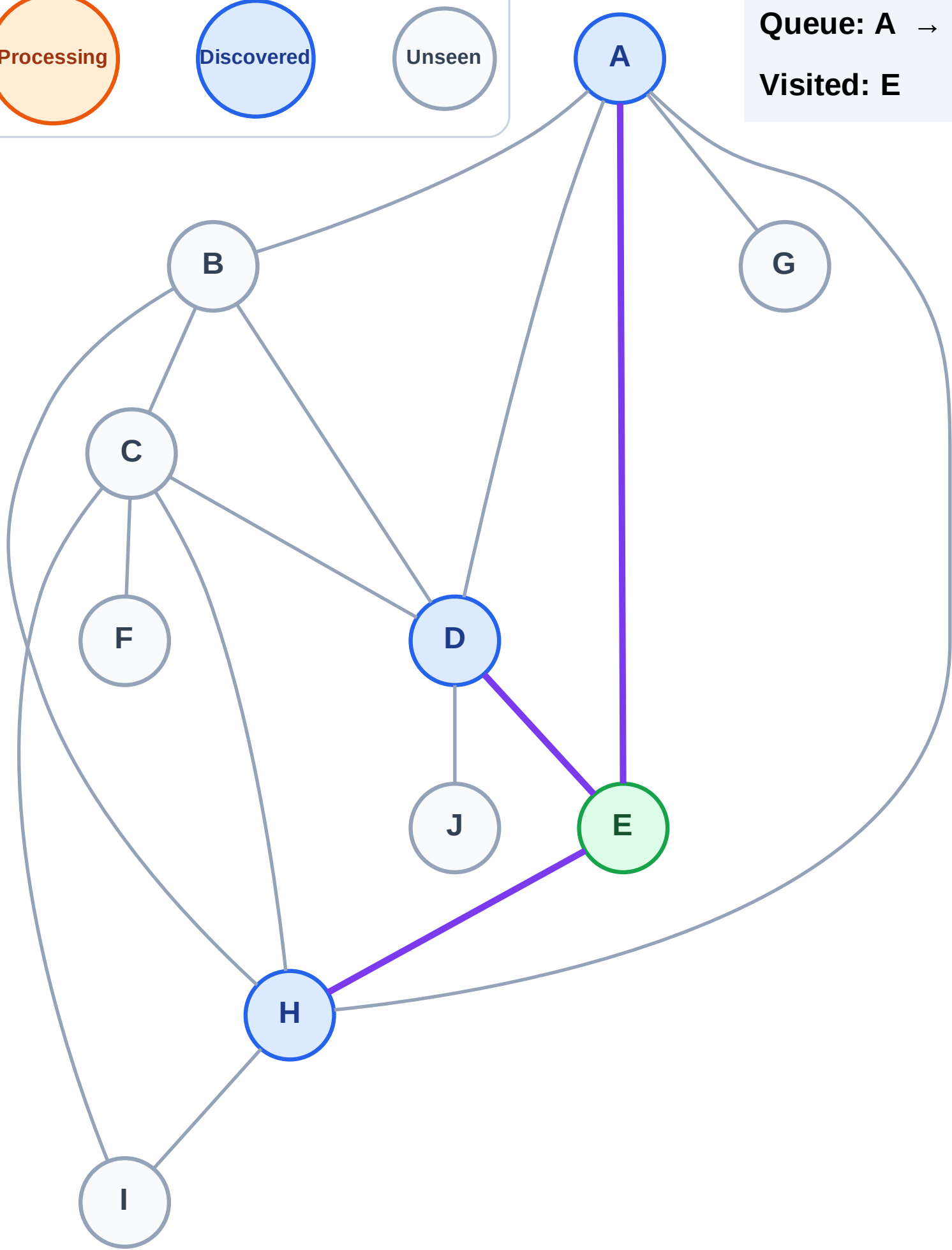
Step 3: Discover A, D, H from E

Legend



Queue: A → D → H

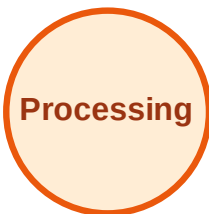
Visited: E



BFS Traversal

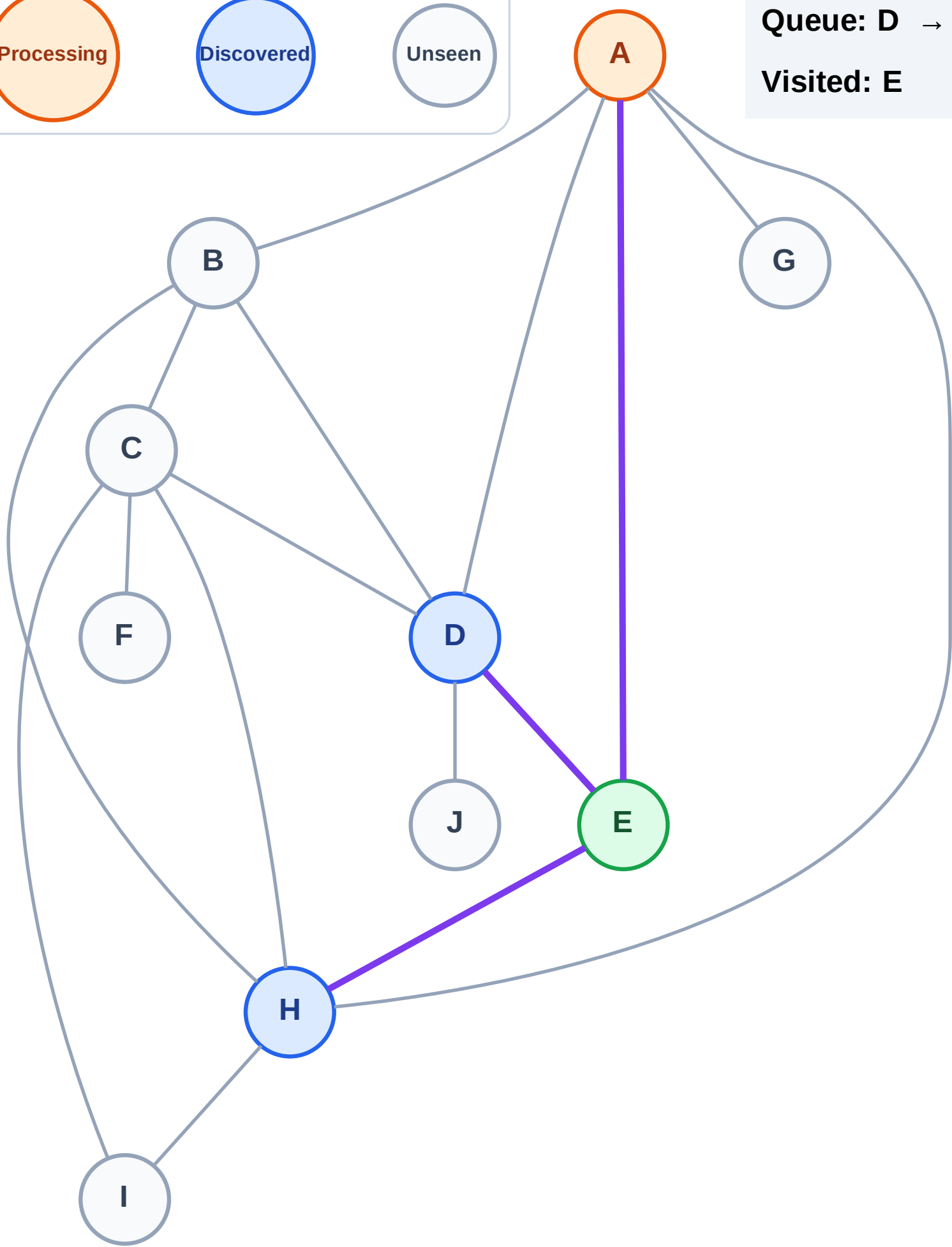
Step 4: Process node A

Legend



Queue: D → H

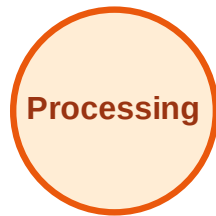
Visited: E



BFS Traversal

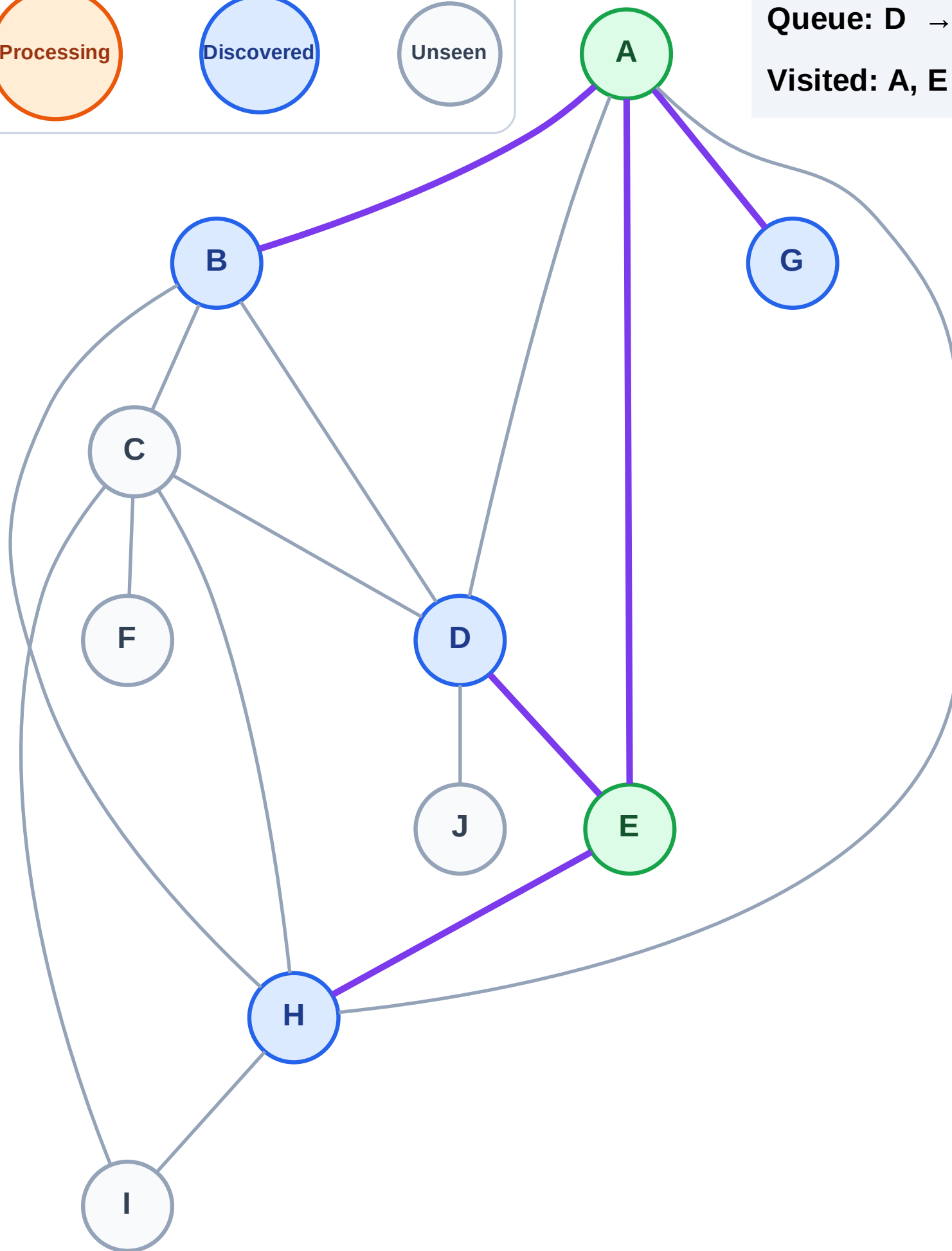
Step 5: Discover B, G from A

Legend



Queue: D → H → B → G

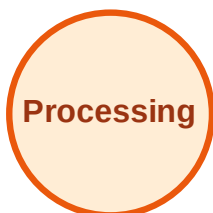
Visited: A, E



BFS Traversal

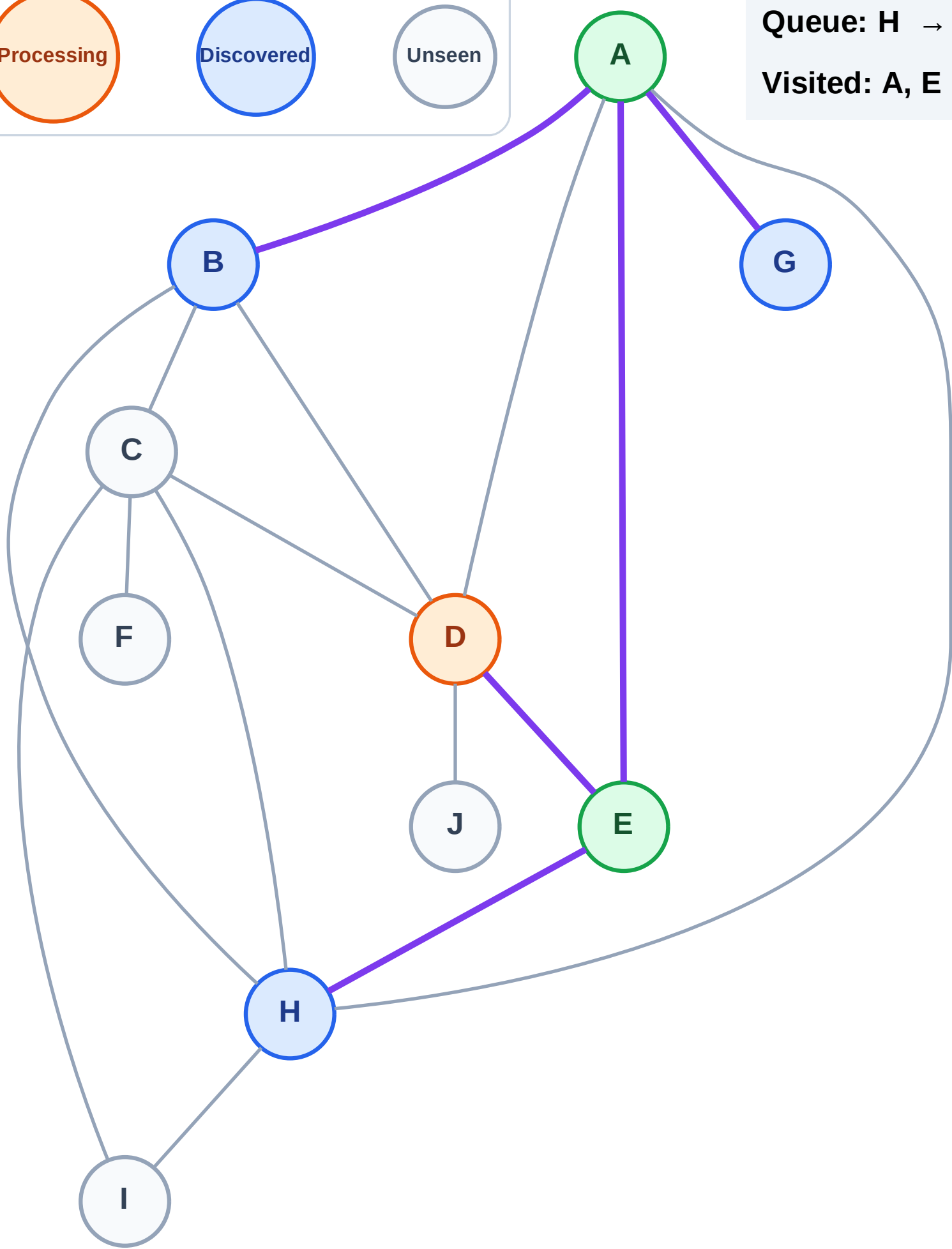
Step 6: Process node D

Legend



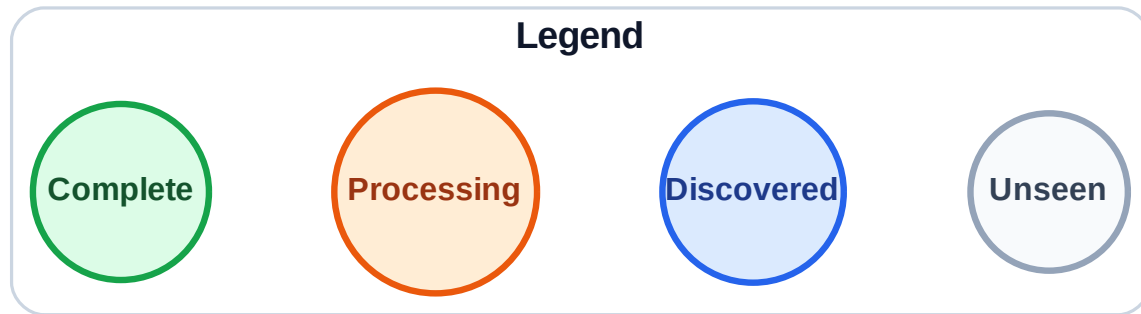
Queue: H → B → G

Visited: A, E



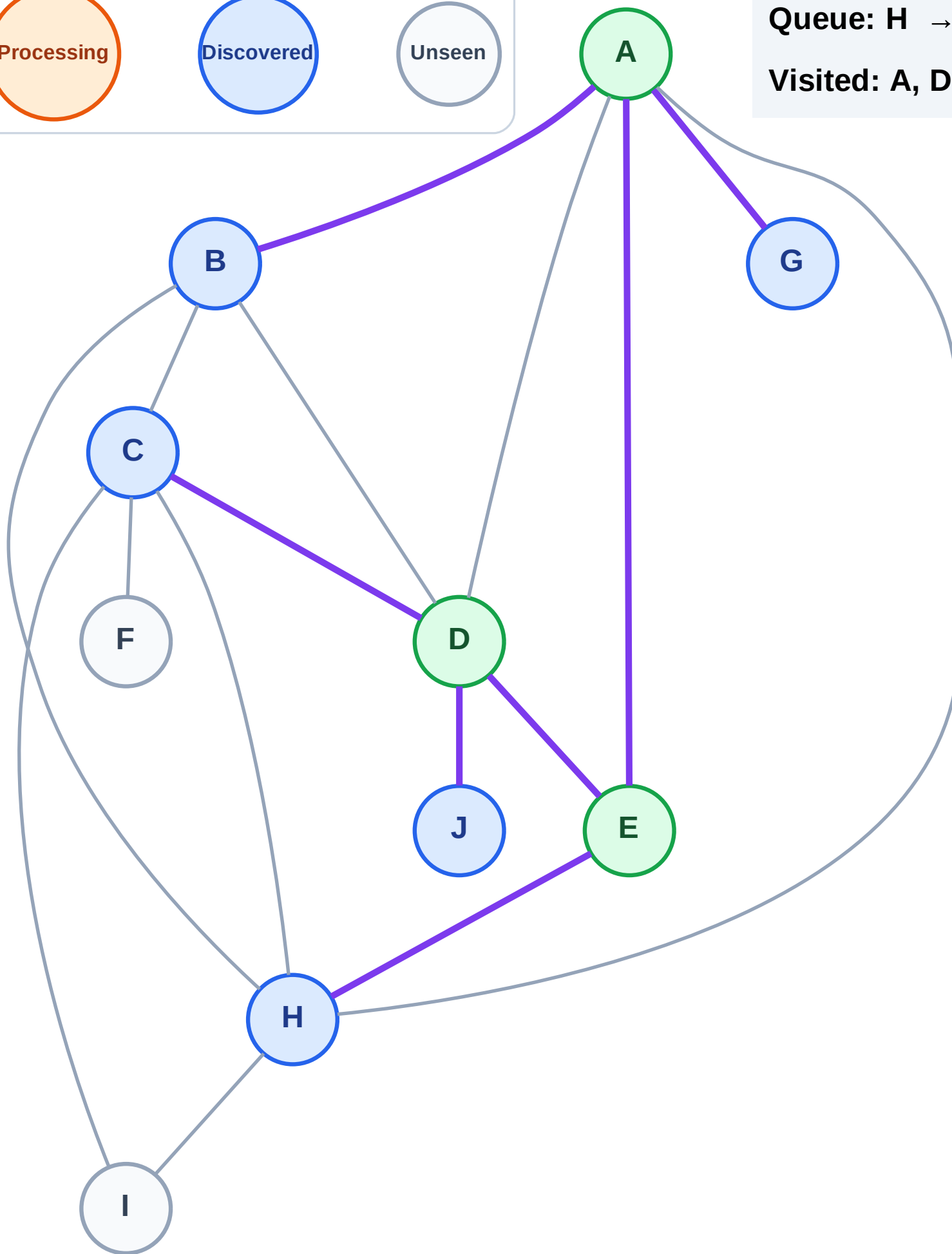
BFS Traversal

Step 7: Discover C, J from D



Queue: H → B → G → C → J

Visited: A, D, E



BFS Traversal

Step 8: Process node H

Legend

Complete

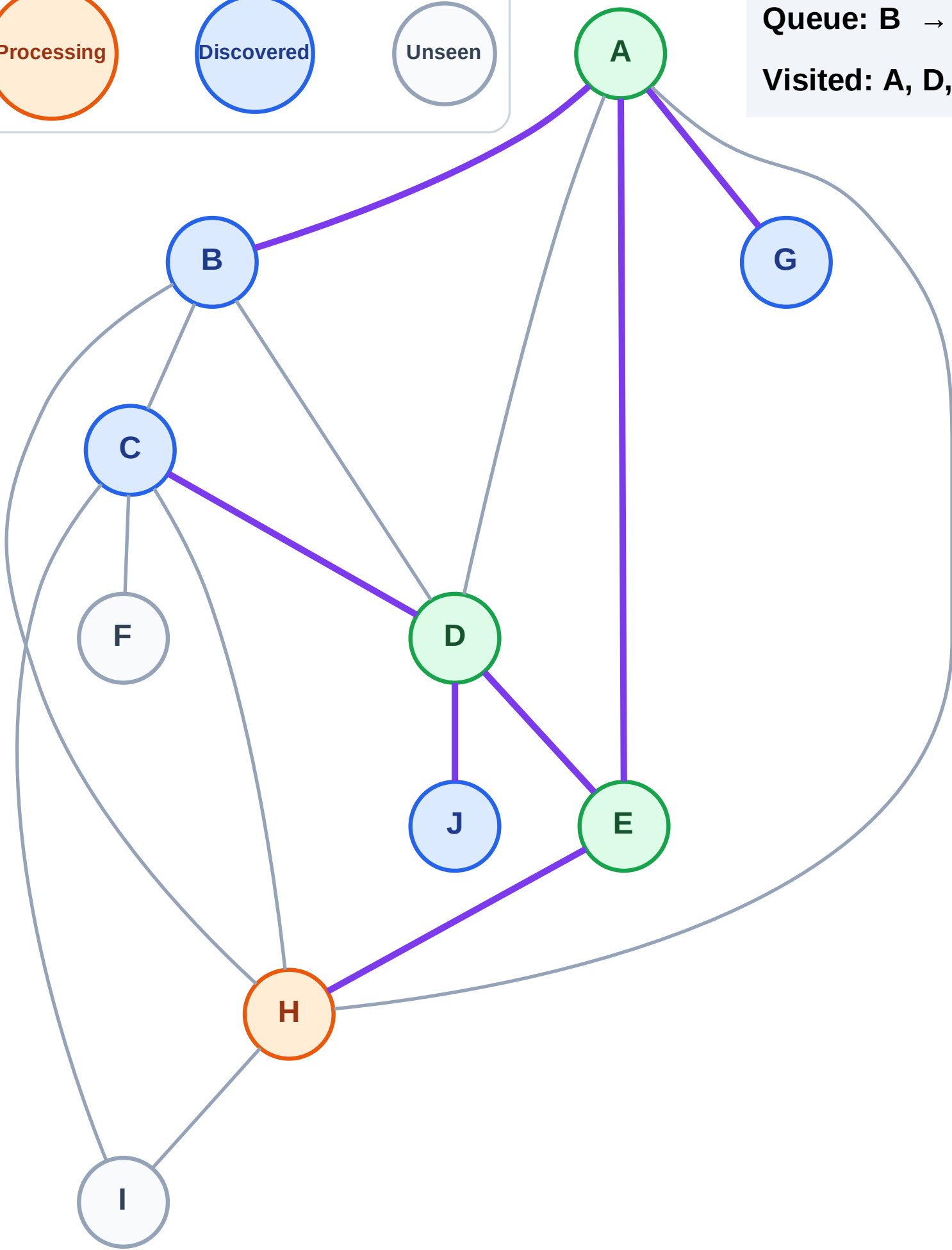
Processing

Discovered

Unseen

Queue: B → G → C → J

Visited: A, D, E



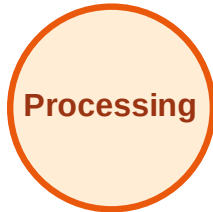
BFS Traversal

Step 9: Discover I from H

Legend



Complete



Processing



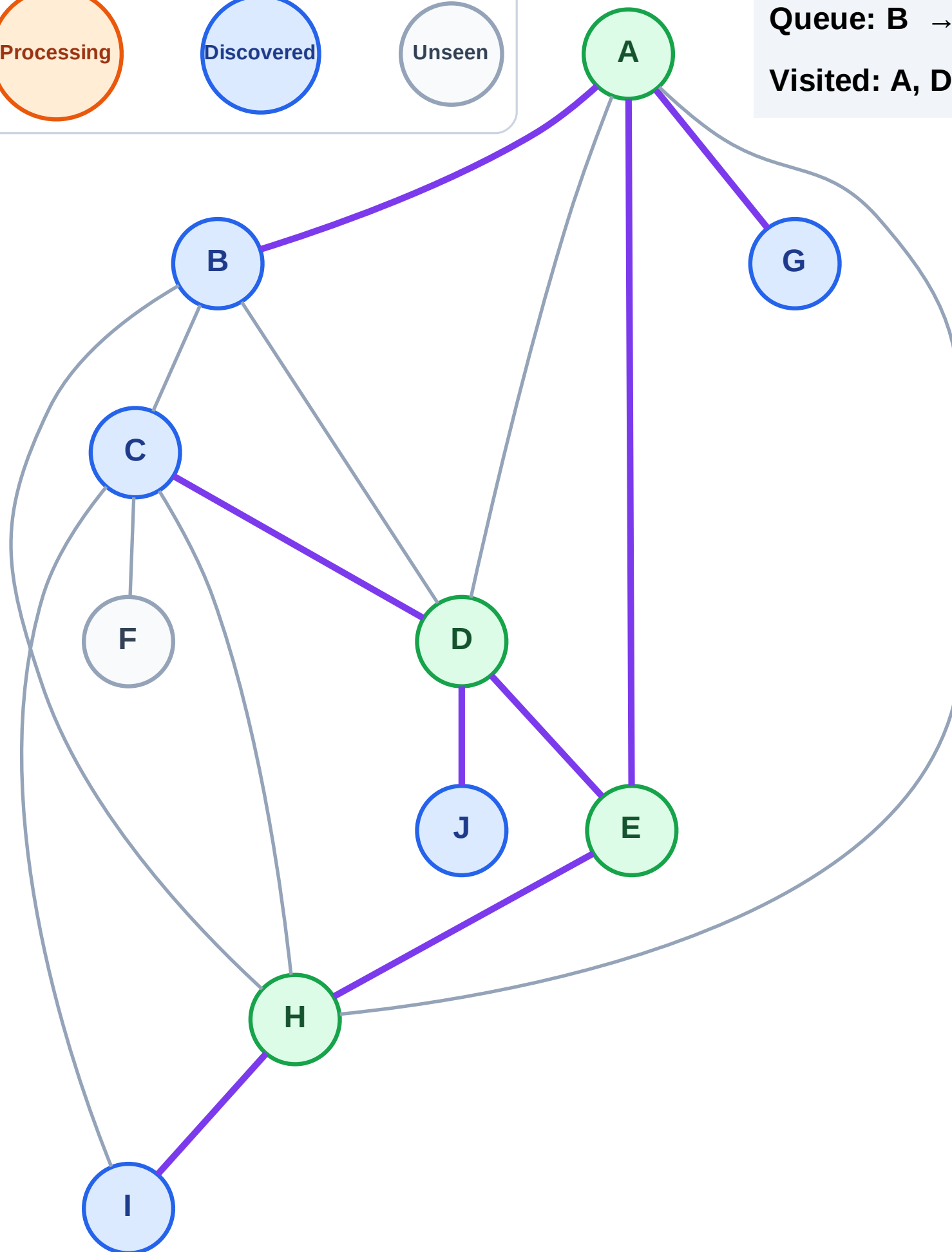
Discovered



Unseen

Queue: B → G → C → J → I

Visited: A, D, E, H



BFS Traversal

Step 10: Process node B

Legend

Complete

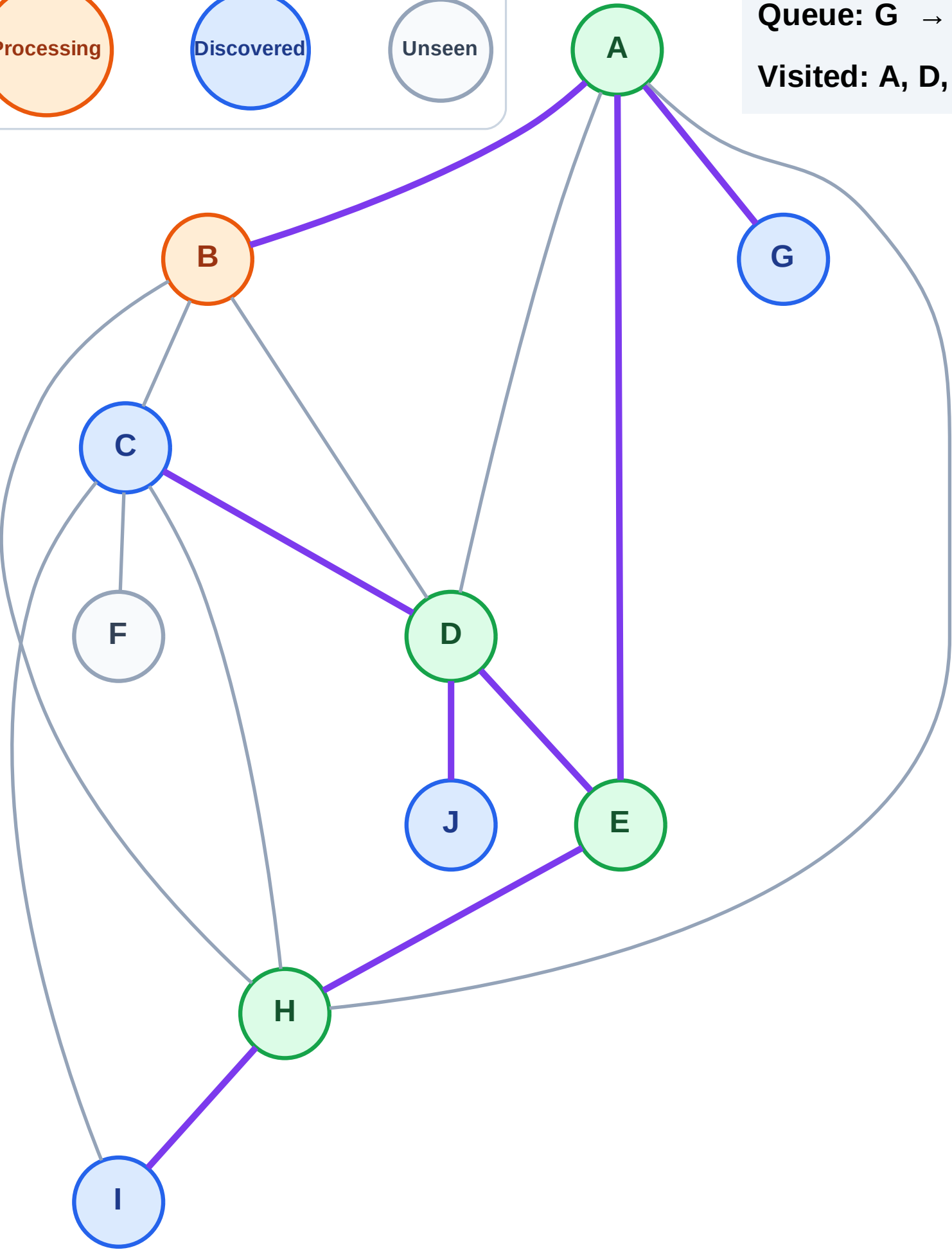
Processing

Discovered

Unseen

Queue: G → C → J → I

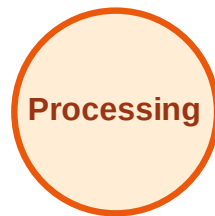
Visited: A, D, E, H



BFS Traversal

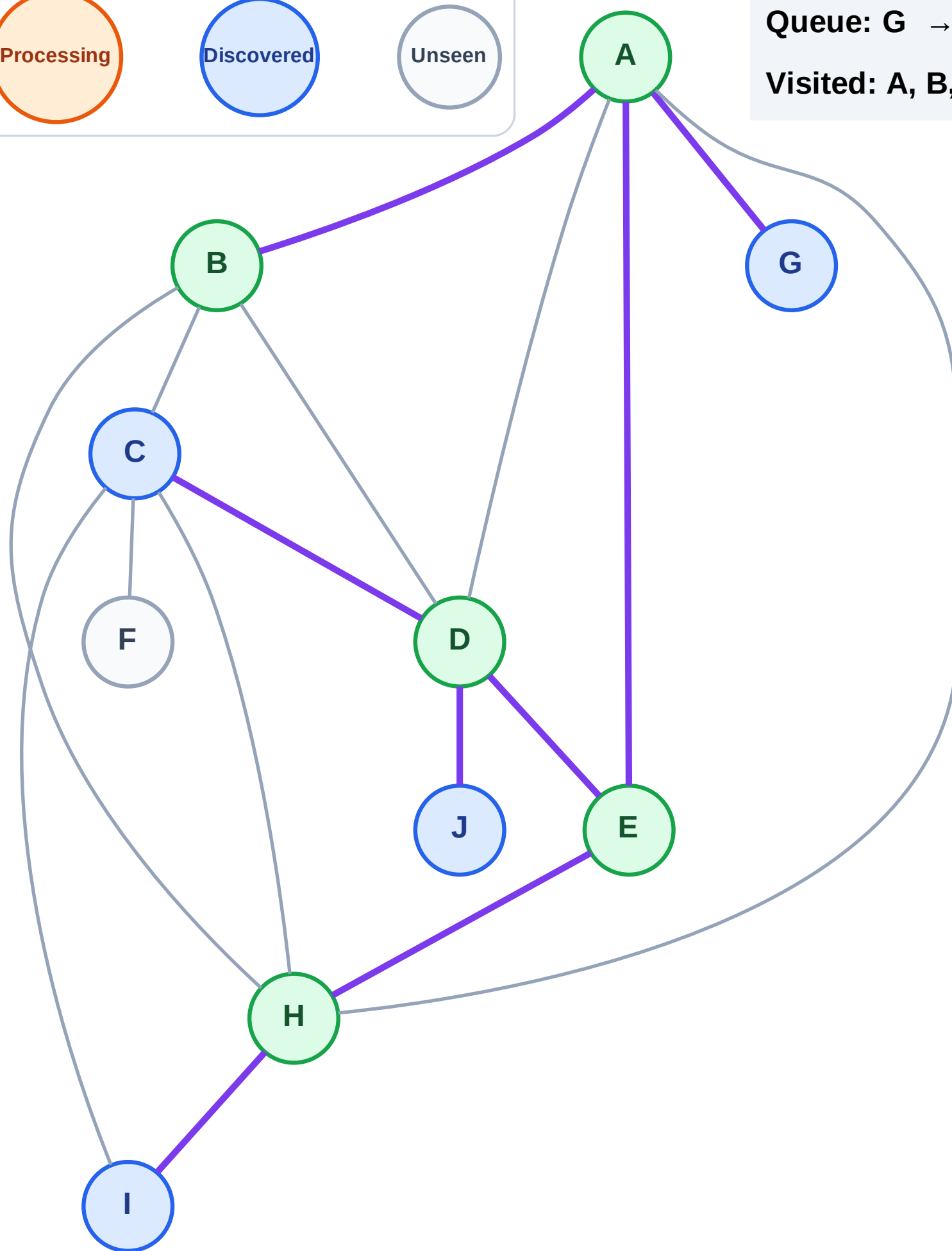
Step 11: No new neighbors from B

Legend



Queue: G → C → J → I

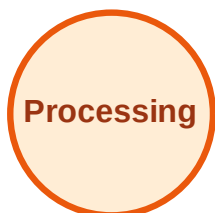
Visited: A, B, D, E, H



BFS Traversal

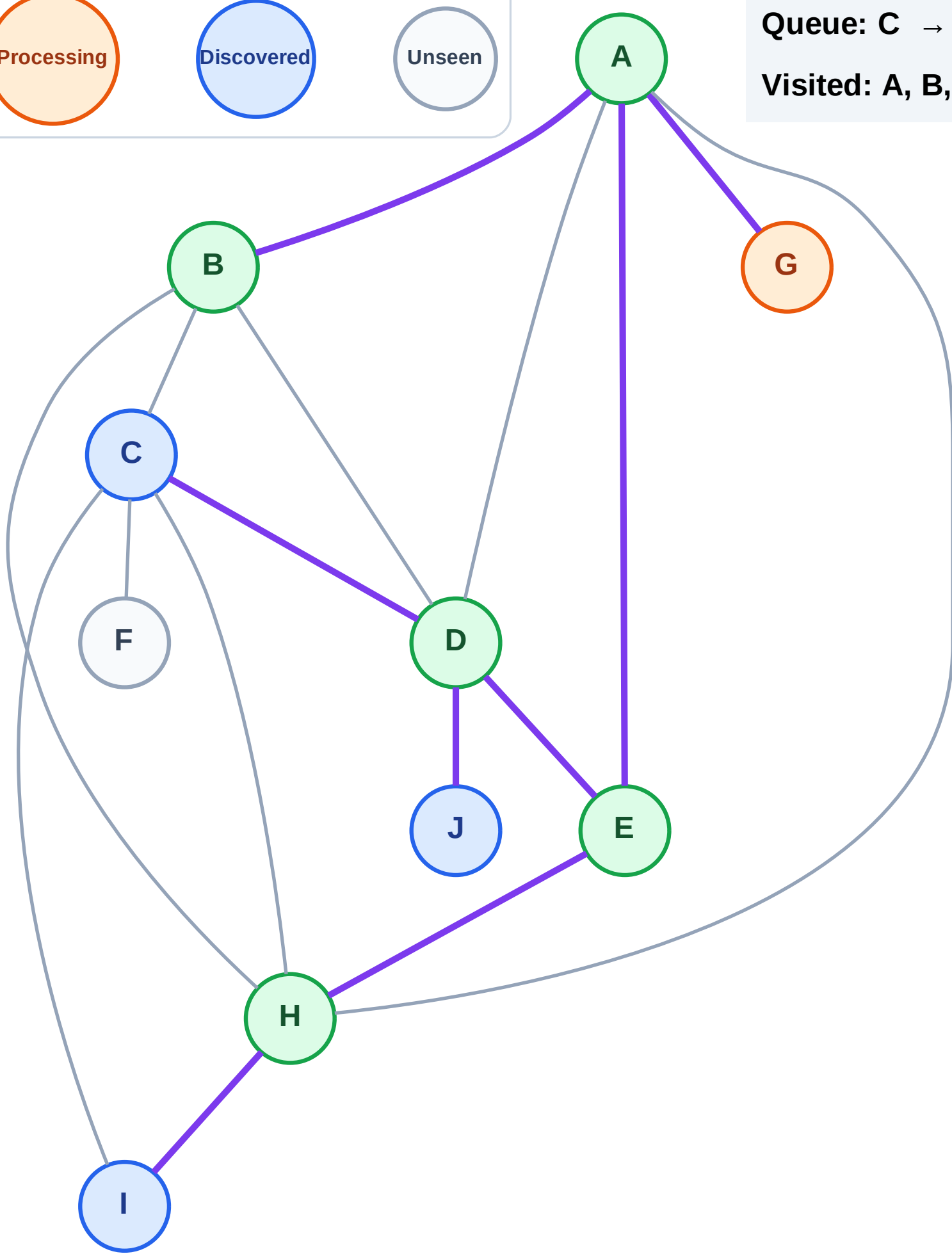
Step 12: Process node G

Legend



Queue: C → J → I

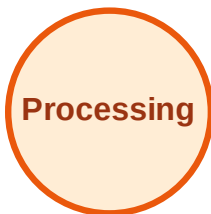
Visited: A, B, D, E, H



BFS Traversal

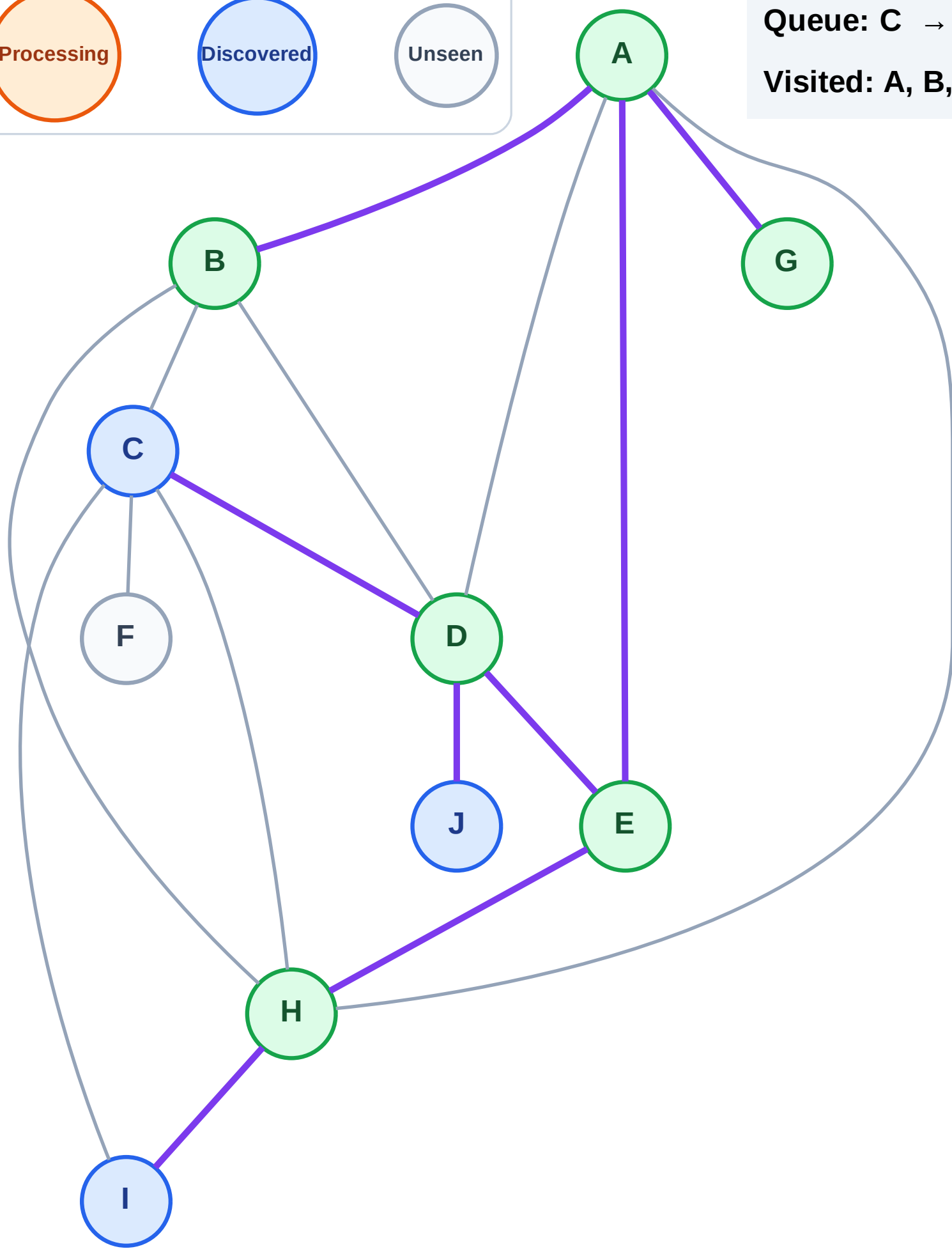
Step 13: No new neighbors from G

Legend



Queue: C → J → I

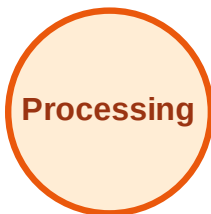
Visited: A, B, D, E, G, H



BFS Traversal

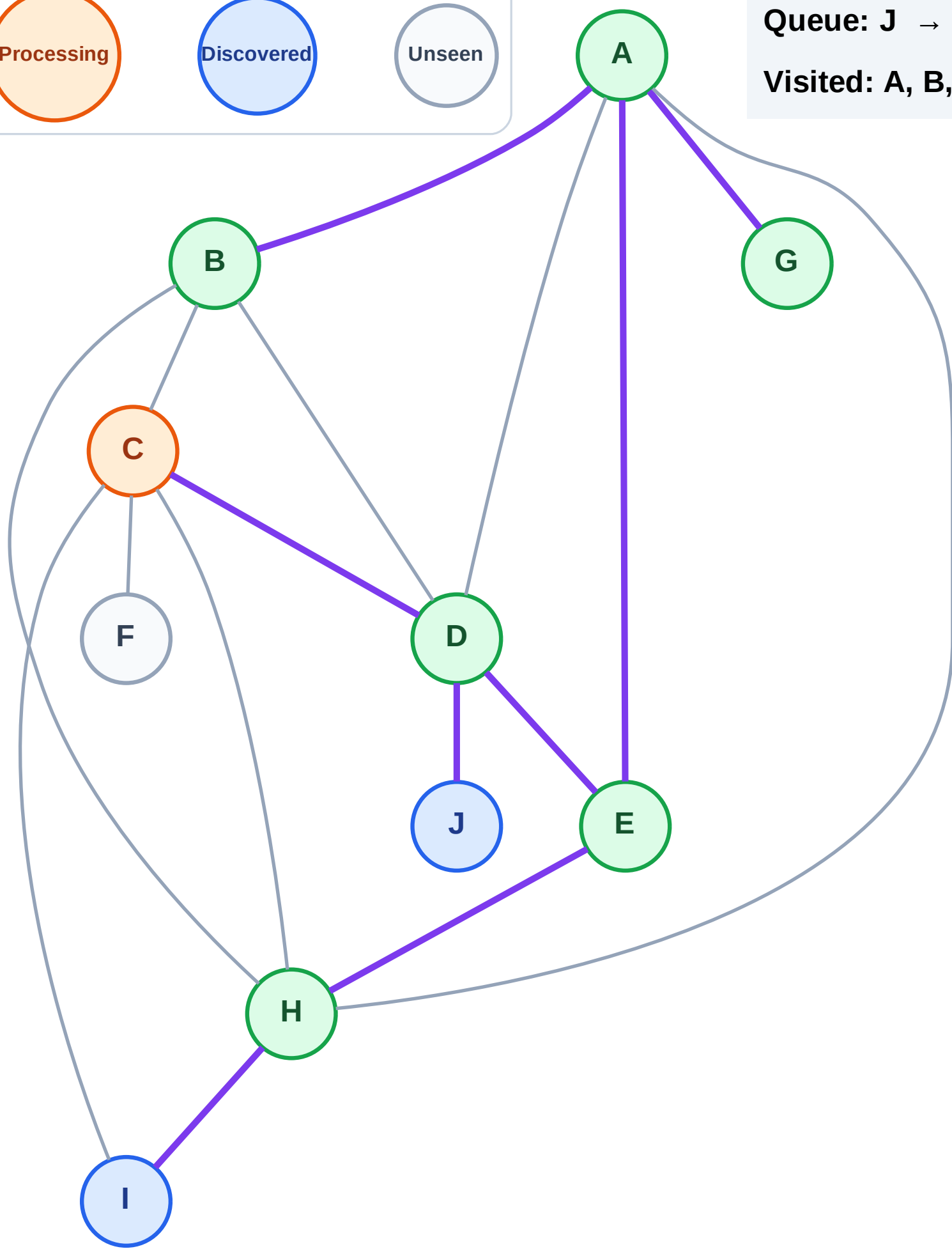
Step 14: Process node C

Legend



Queue: J → I

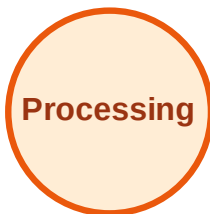
Visited: A, B, D, E, G, H



BFS Traversal

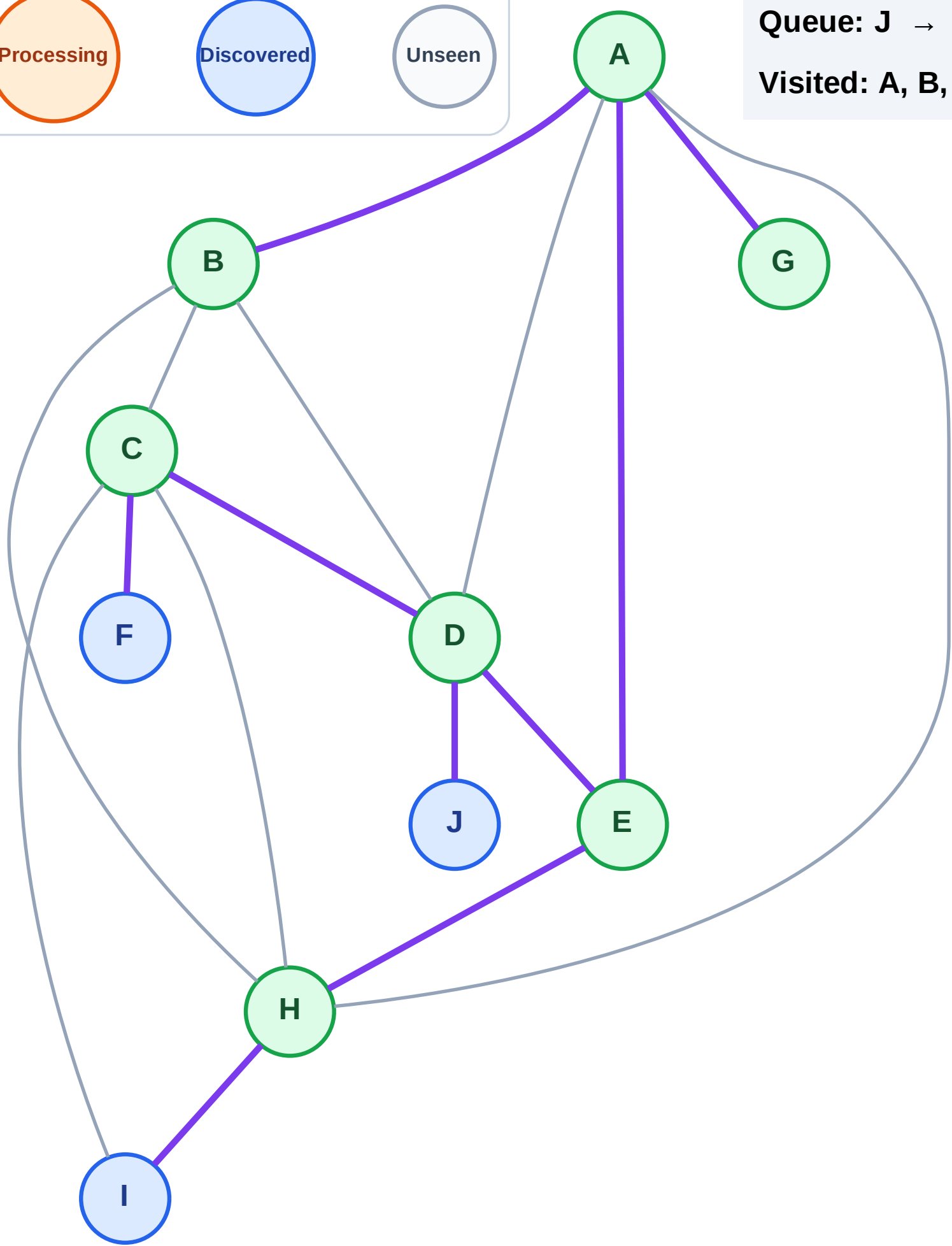
Step 15: Discover F from C

Legend



Queue: J → I → F

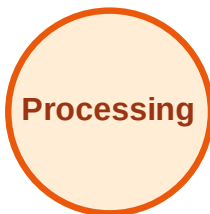
Visited: A, B, C, D, E, G, H



BFS Traversal

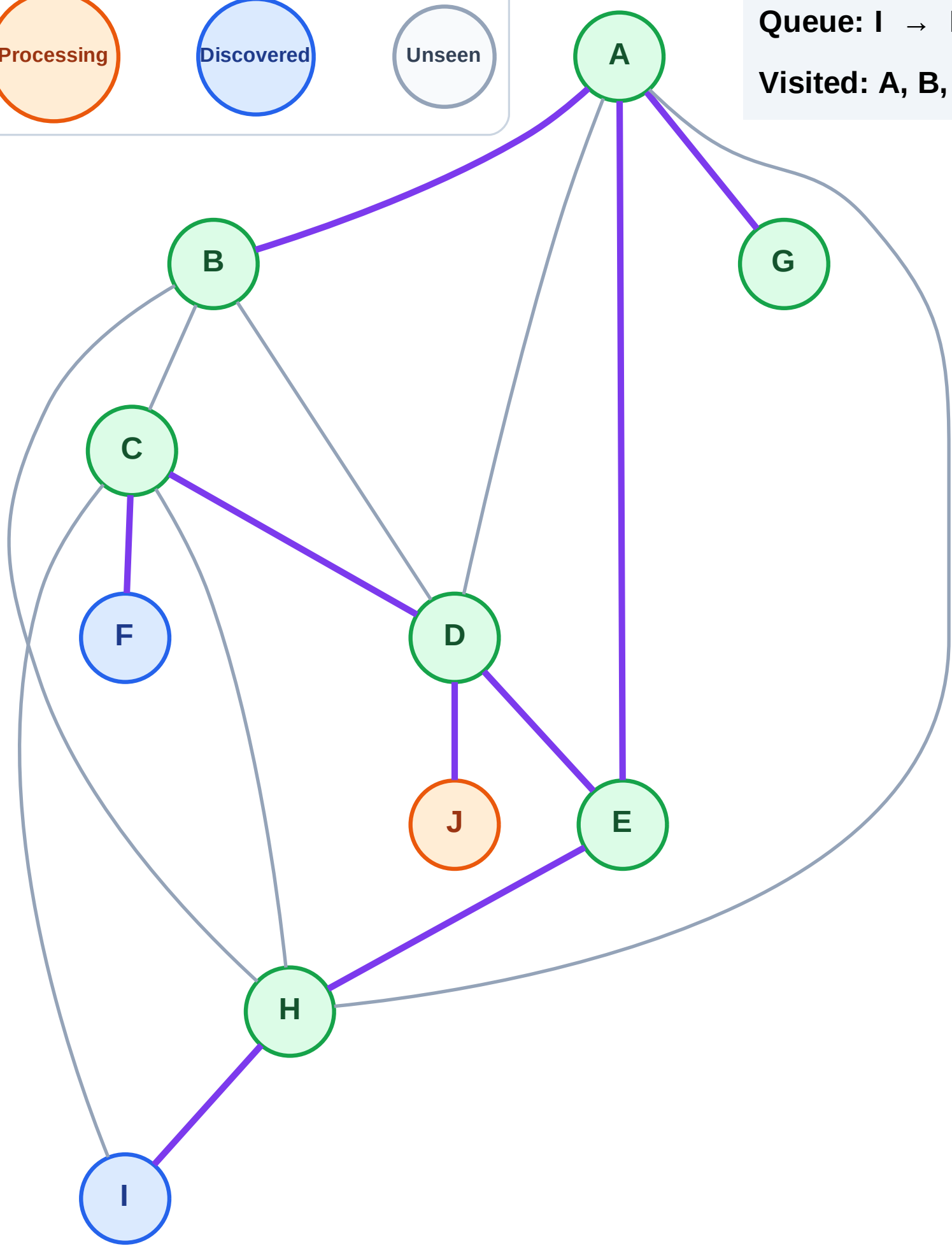
Step 16: Process node J

Legend



Queue: I → F

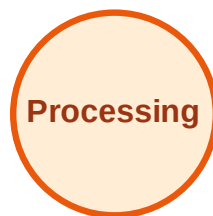
Visited: A, B, C, D, E, G, H



BFS Traversal

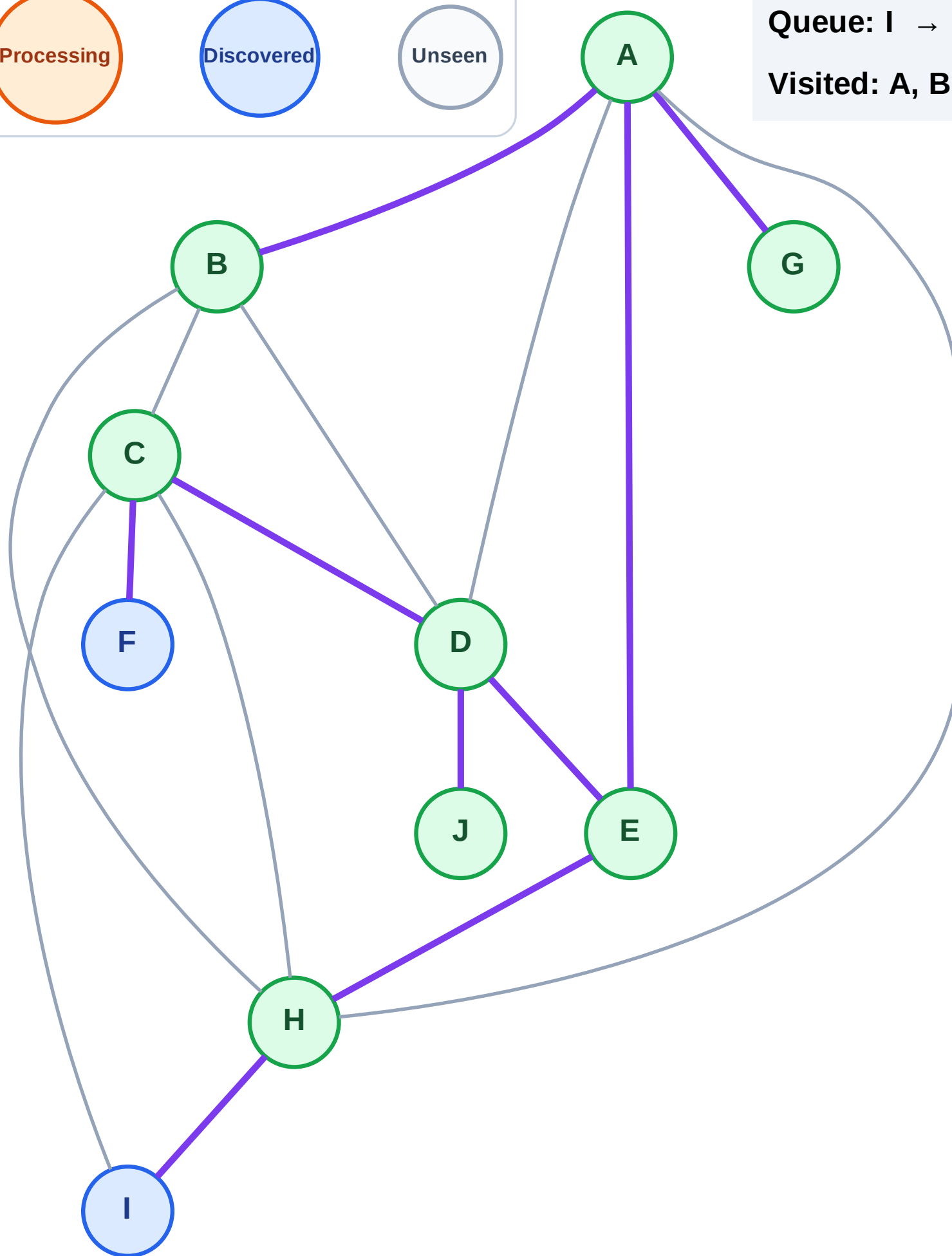
Step 17: No new neighbors from J

Legend



Queue: I → F

Visited: A, B, C, D, E, G, H, J



BFS Traversal

Step 18: Process node I

Legend

Complete

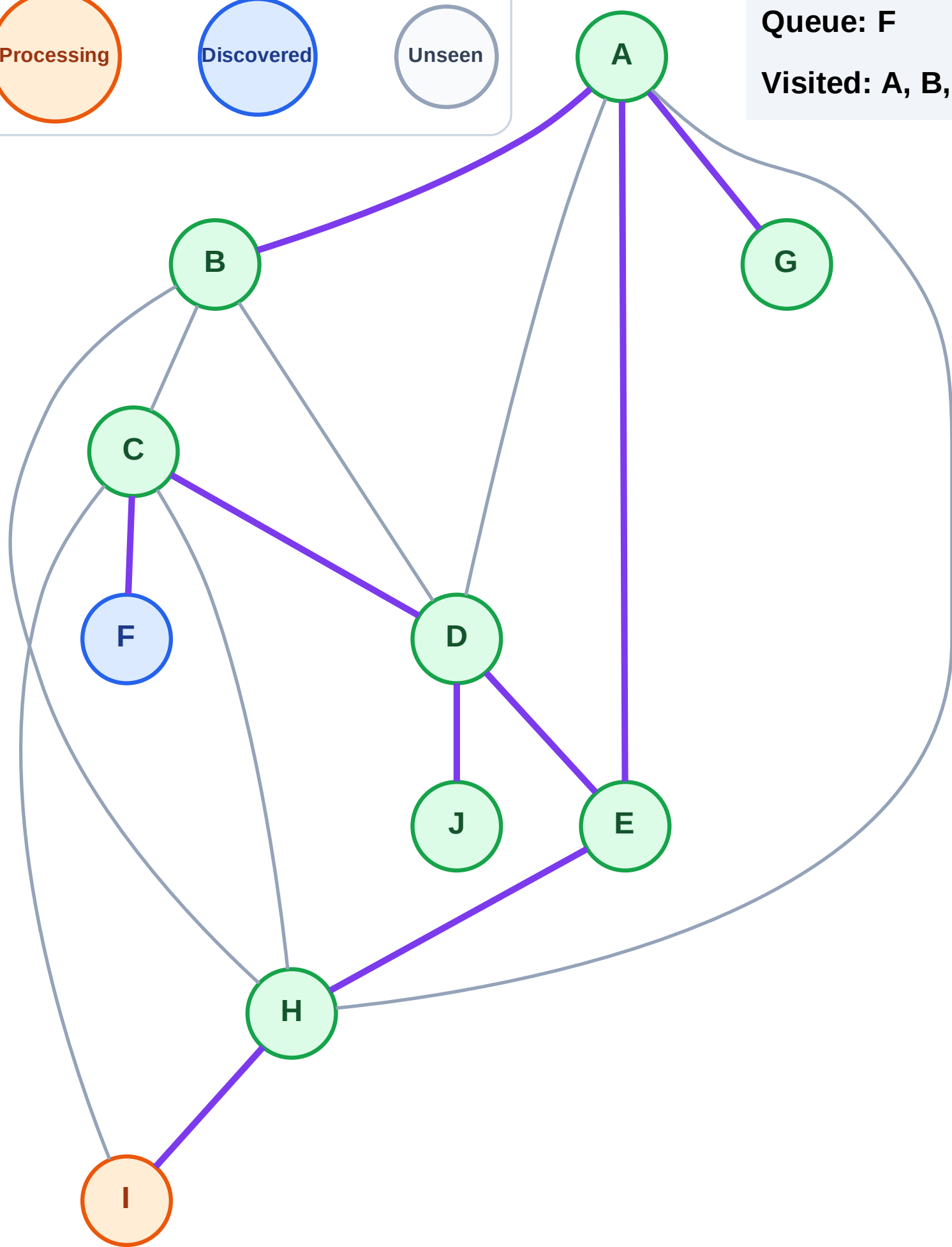
Processing

Discovered

Unseen

Queue: F

Visited: A, B, C, D, E, G, H, J



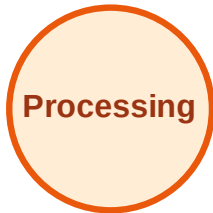
BFS Traversal

Step 19: No new neighbors from I

Legend



Complete



Processing



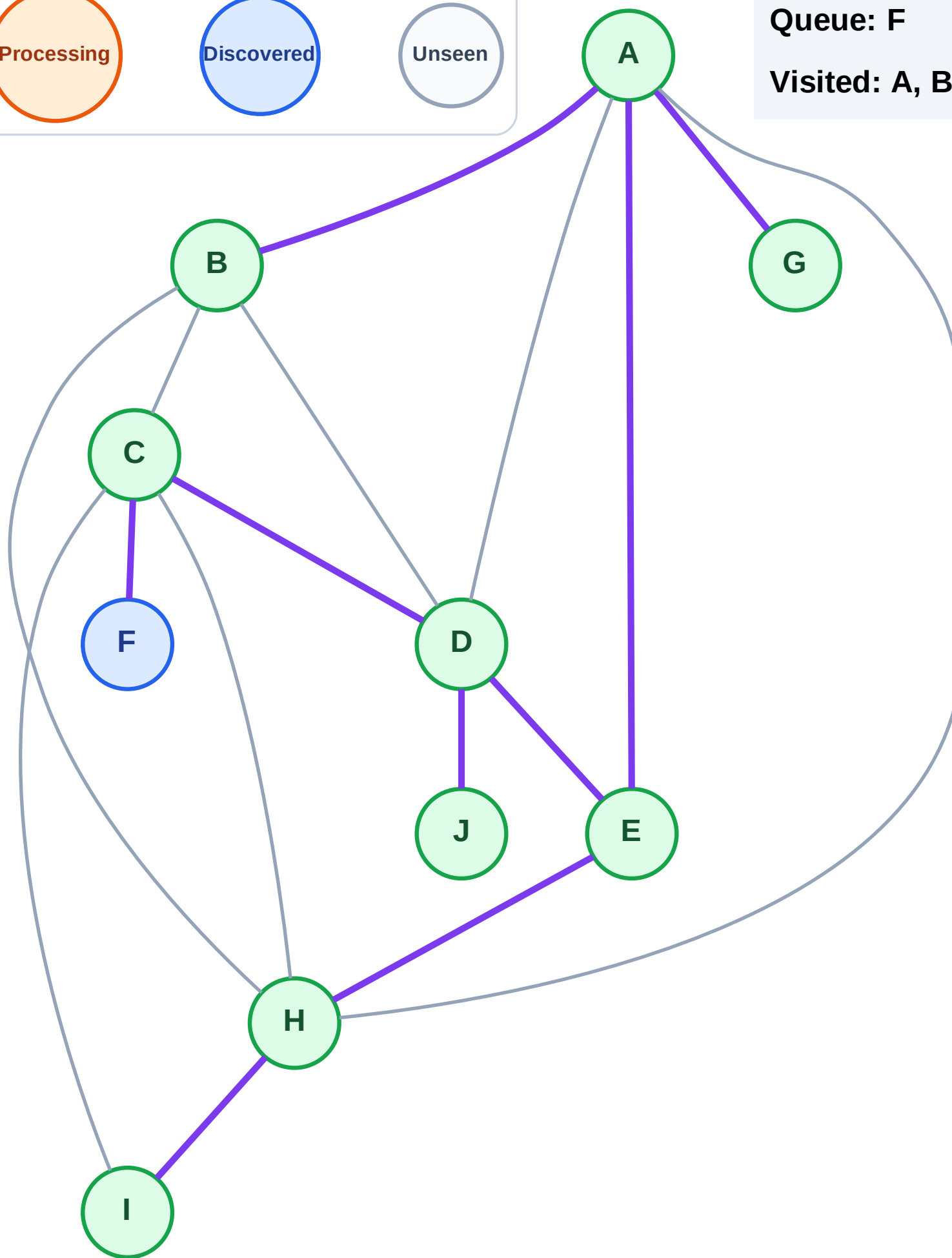
Discovered



Unseen

Queue: F

Visited: A, B, C, D, E, G, H, I, J



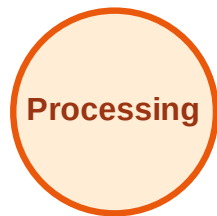
BFS Traversal

Step 20: Process node F

Legend



Complete



Processing



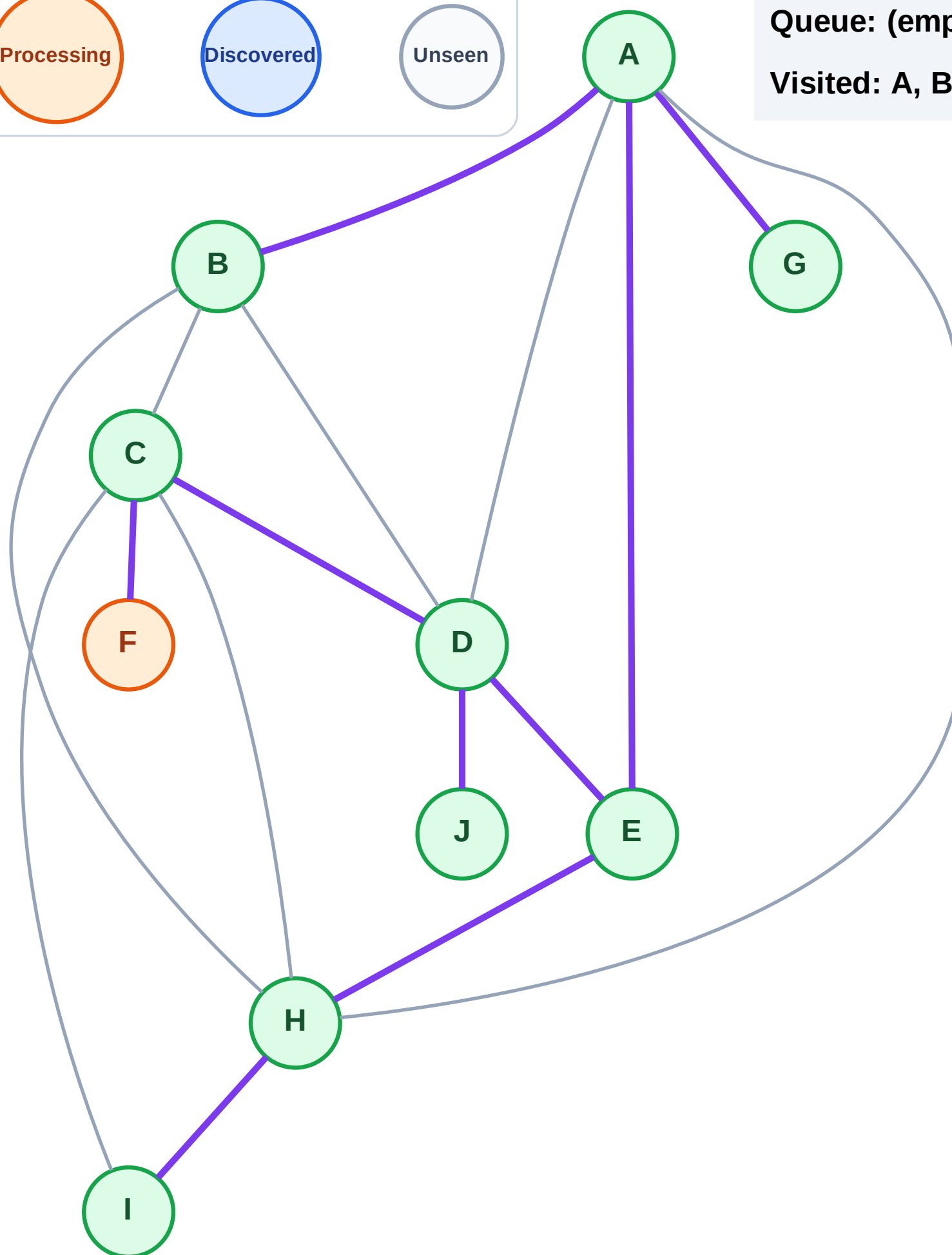
Discovered



Unseen

Queue: (empty)

Visited: A, B, C, D, E, G, H, I, J



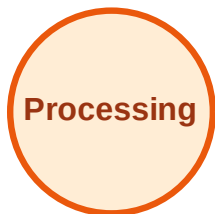
BFS Traversal

Step 21: No new neighbors from F

Legend



Complete



Processing



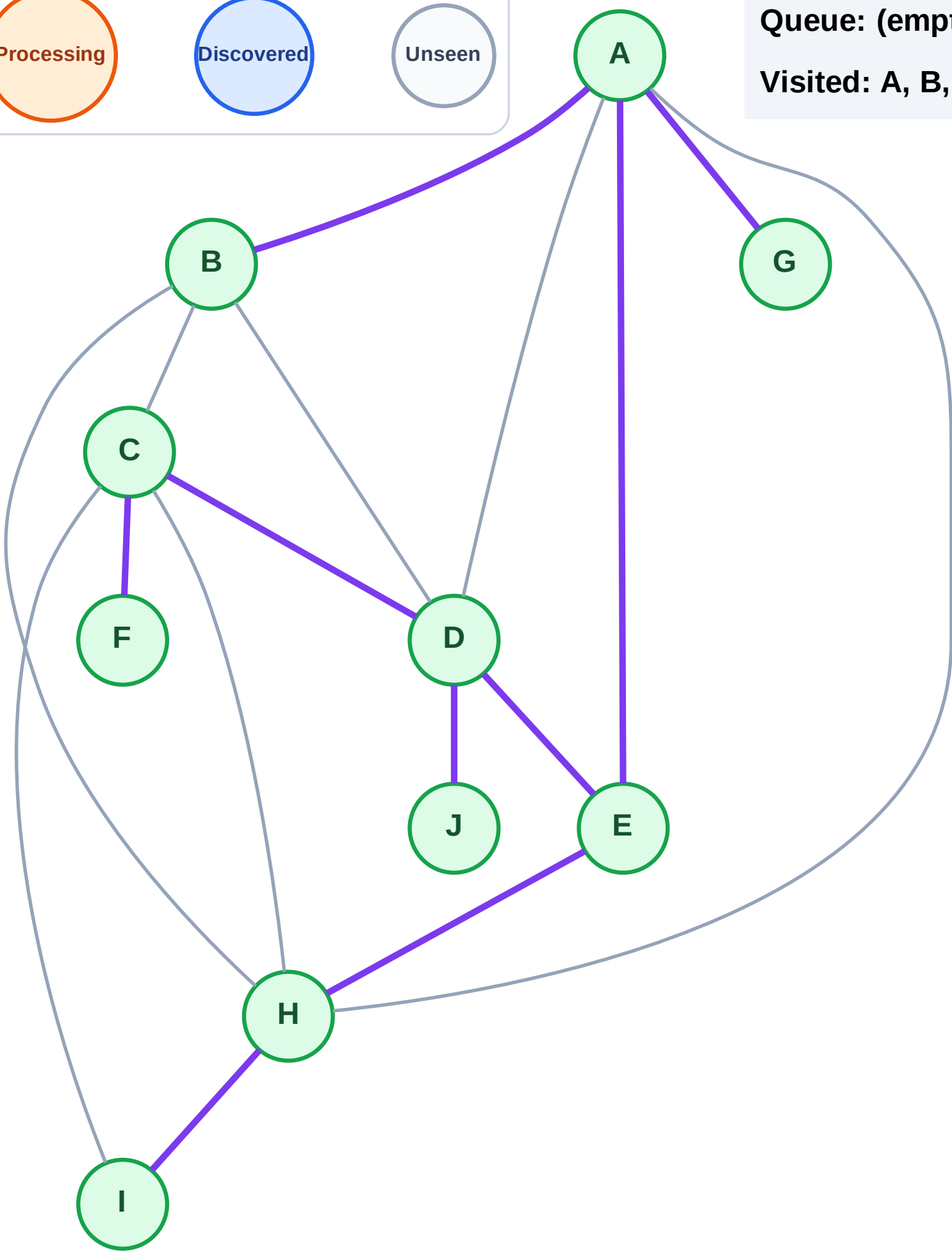
Discovered



Unseen

Queue: (empty)

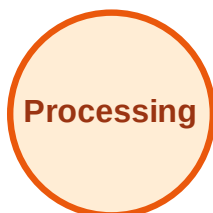
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

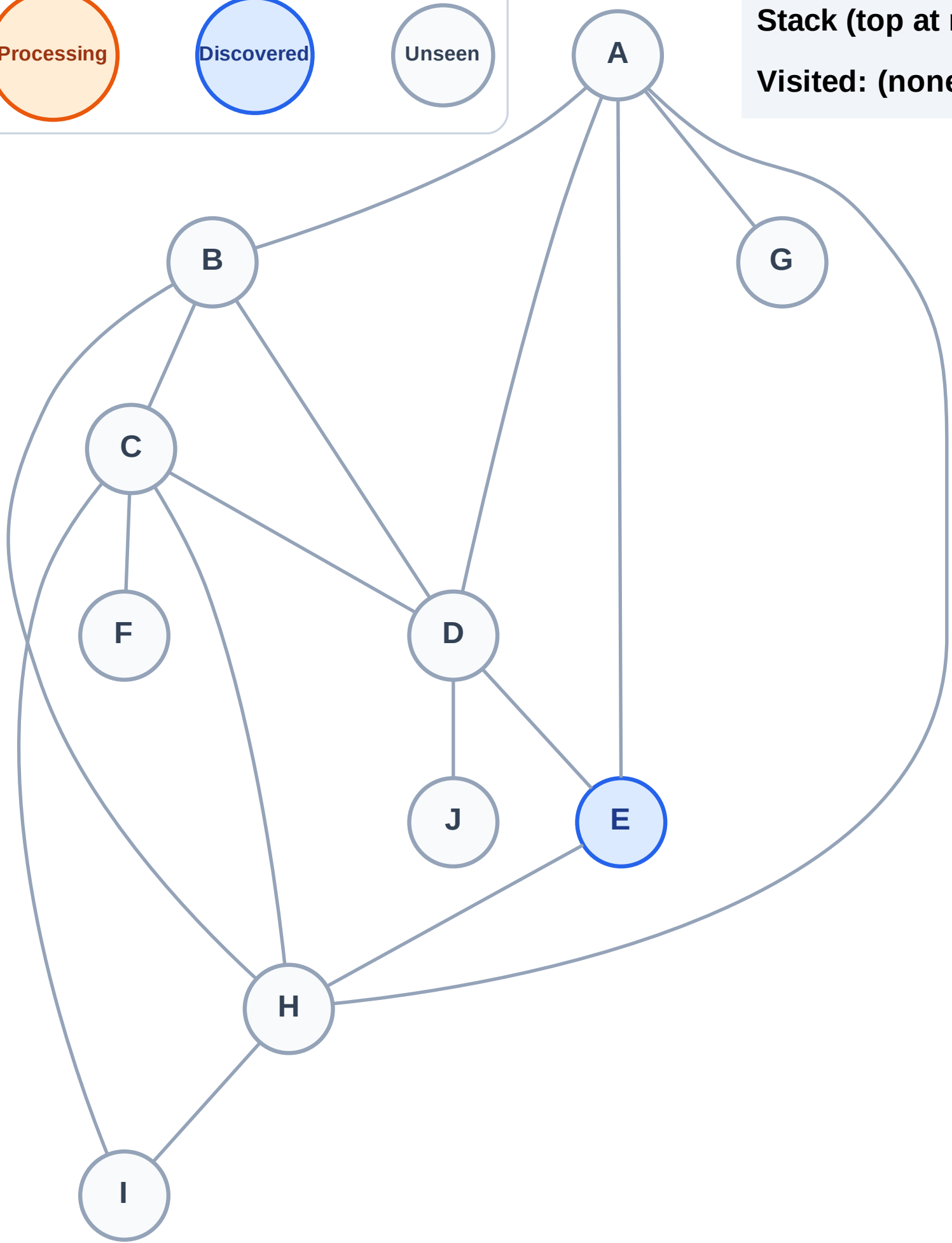
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

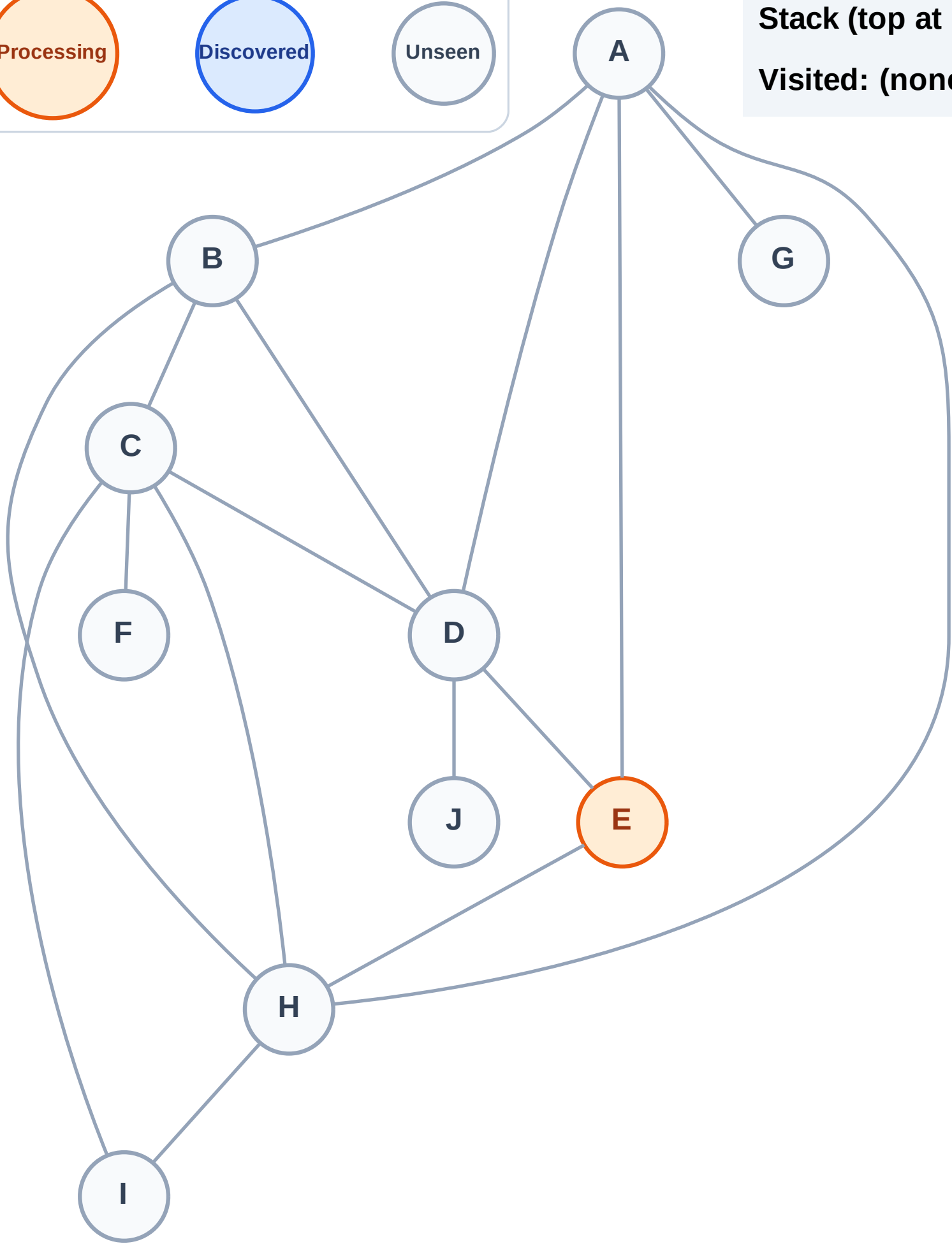
Complete

Processing

Discovered

Unseen

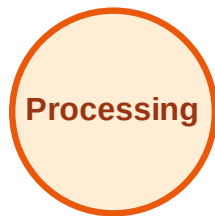
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

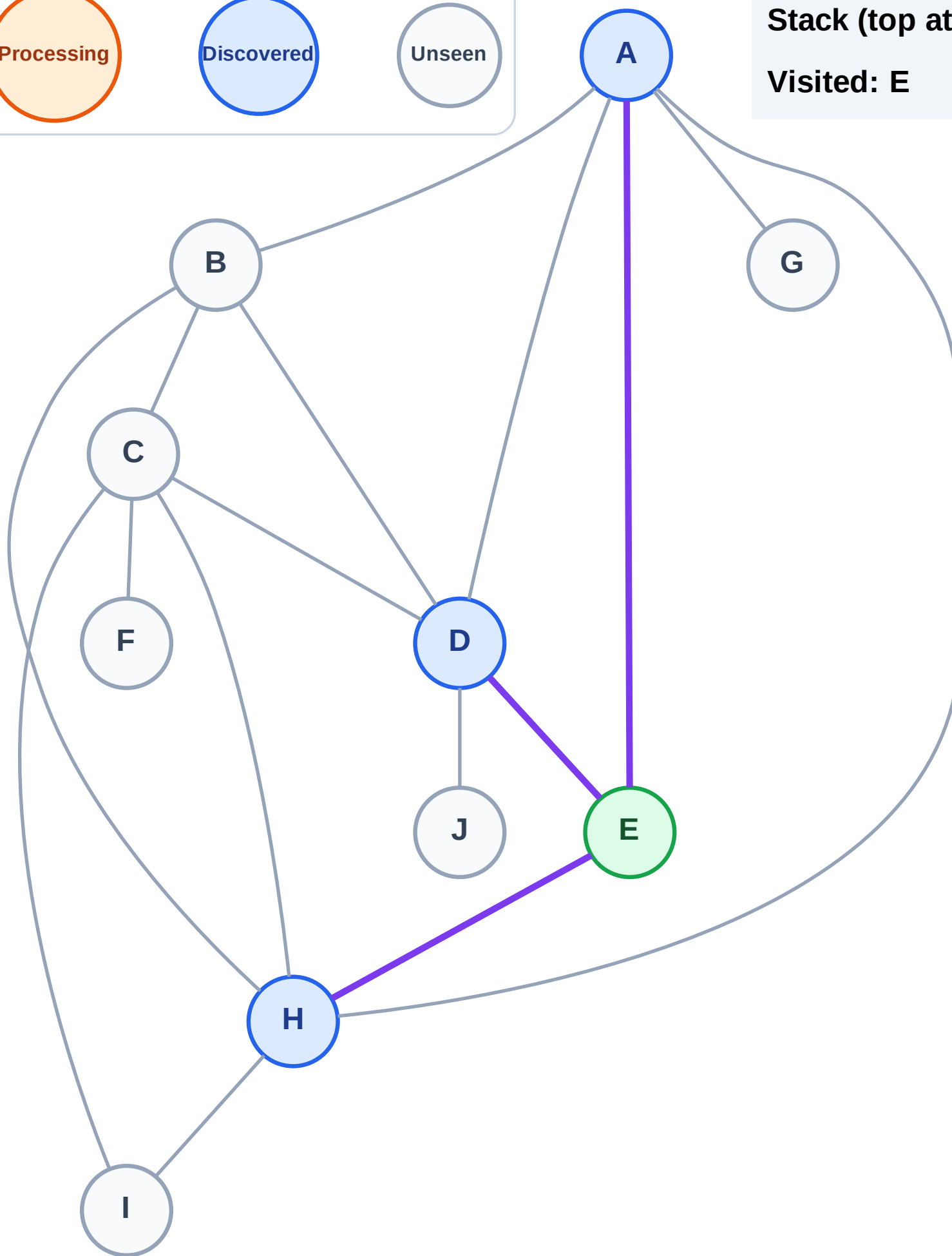
Step 3: Discover H, D, A from E

Legend



Stack (top at right): H | D | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

Complete

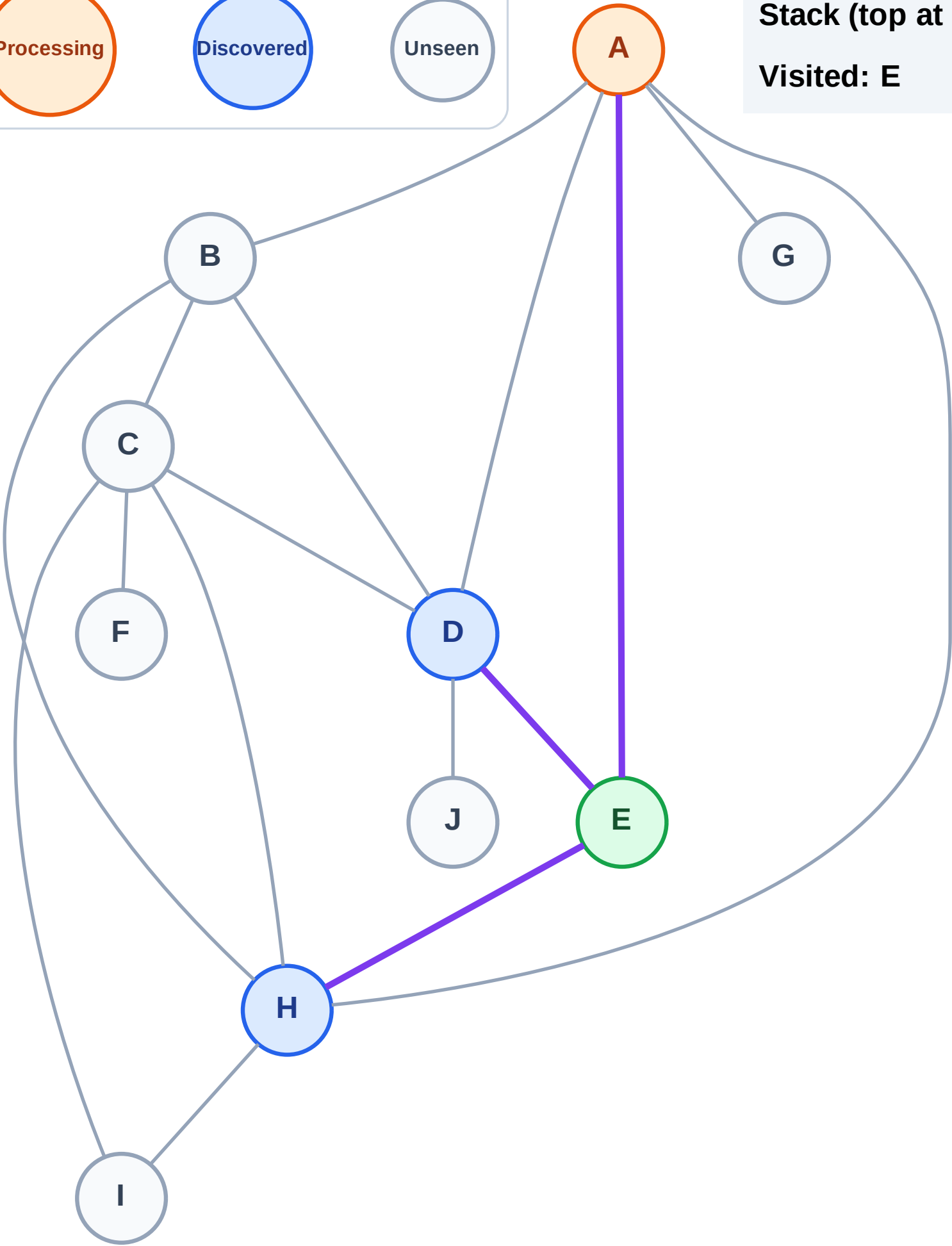
Processing

Discovered

Unseen

Stack (top at right): H | D

Visited: E



DFS Traversal

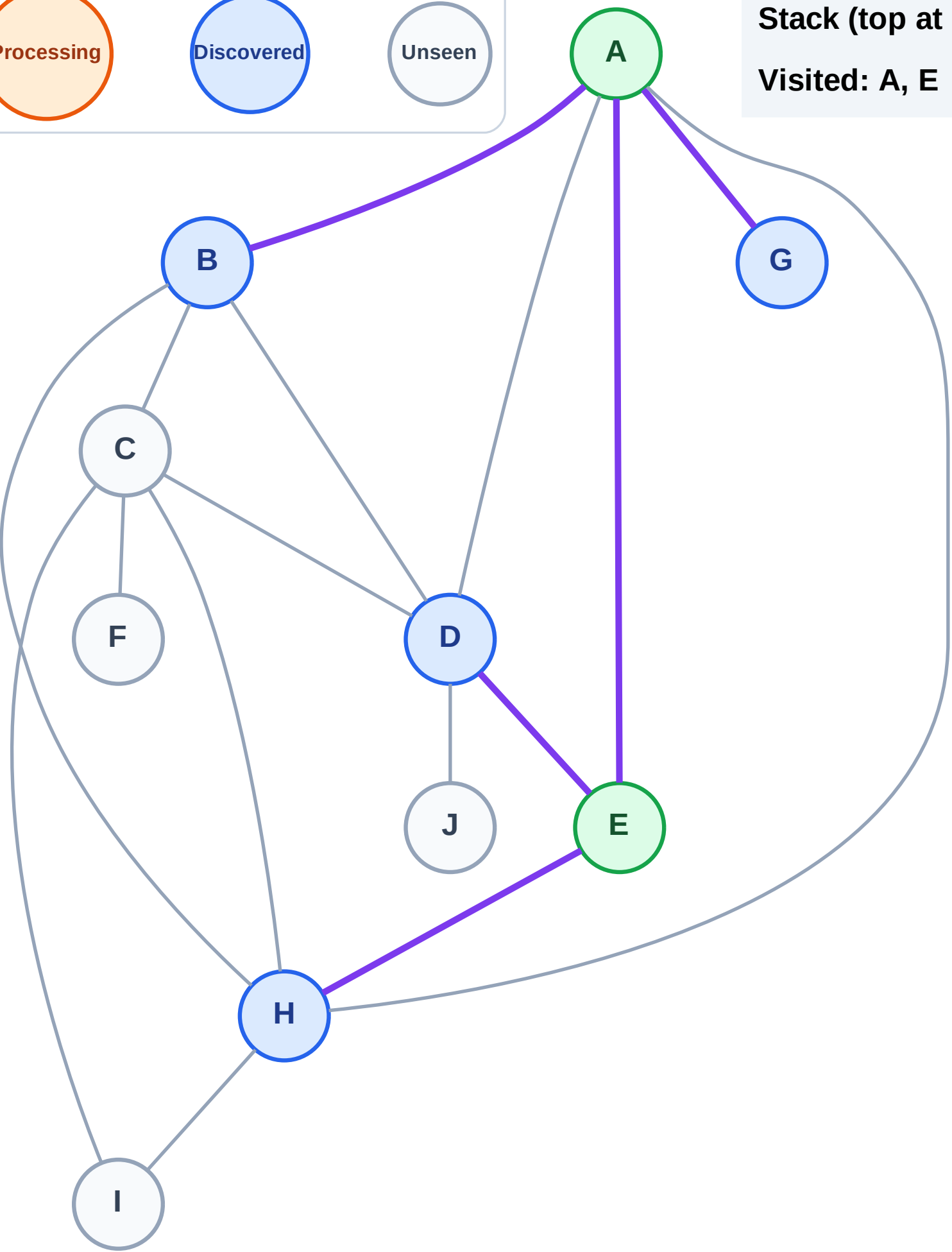
Step 5: Discover G, B from A

Legend



Stack (top at right): H | D | G | B

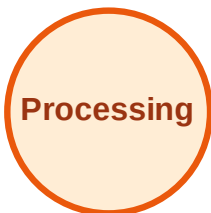
Visited: A, E



DFS Traversal

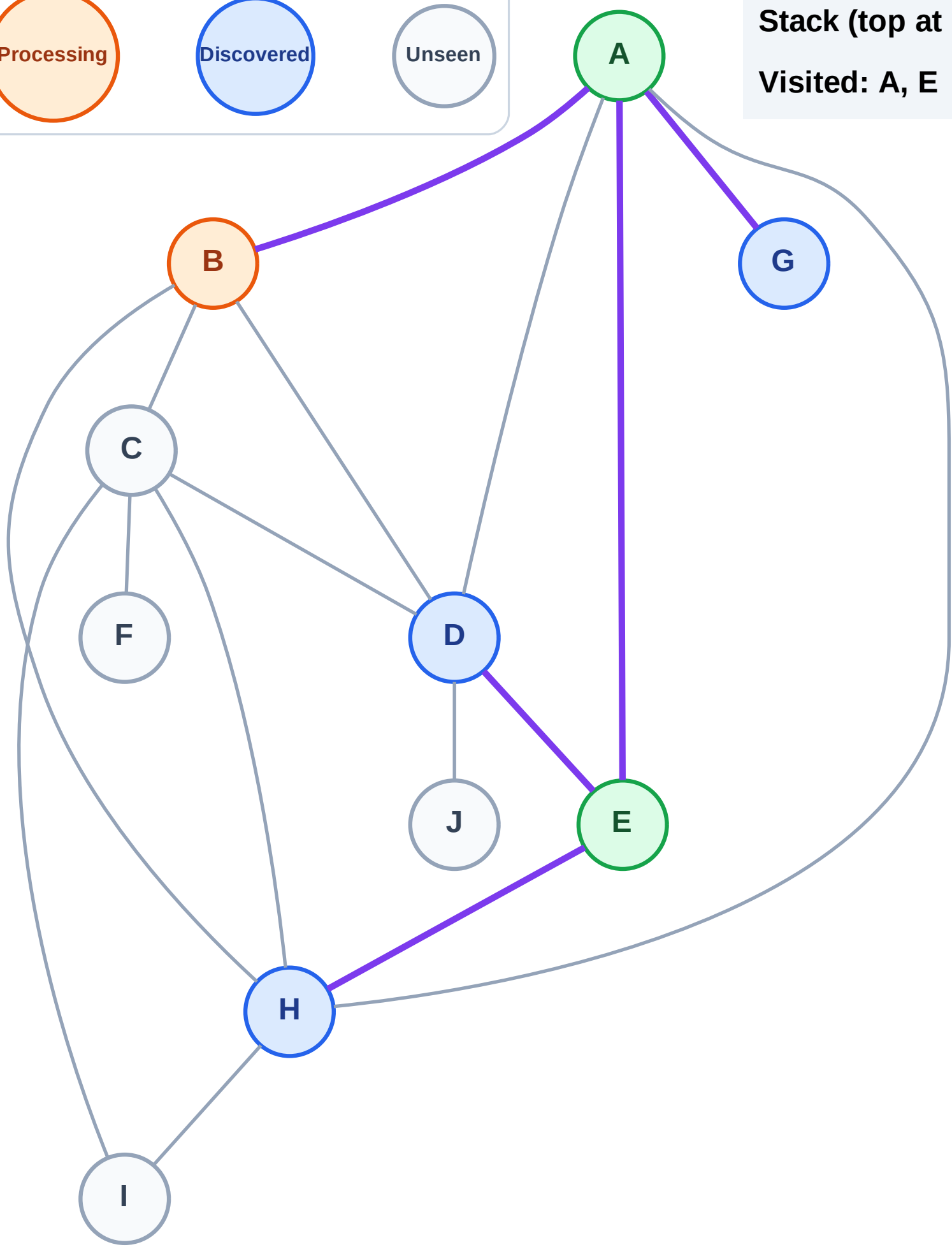
Step 6: Process node B

Legend



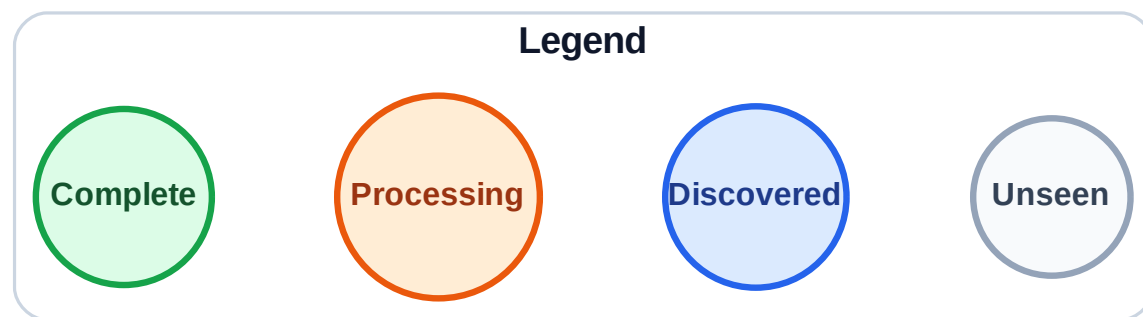
Stack (top at right): H | D | G

Visited: A, E



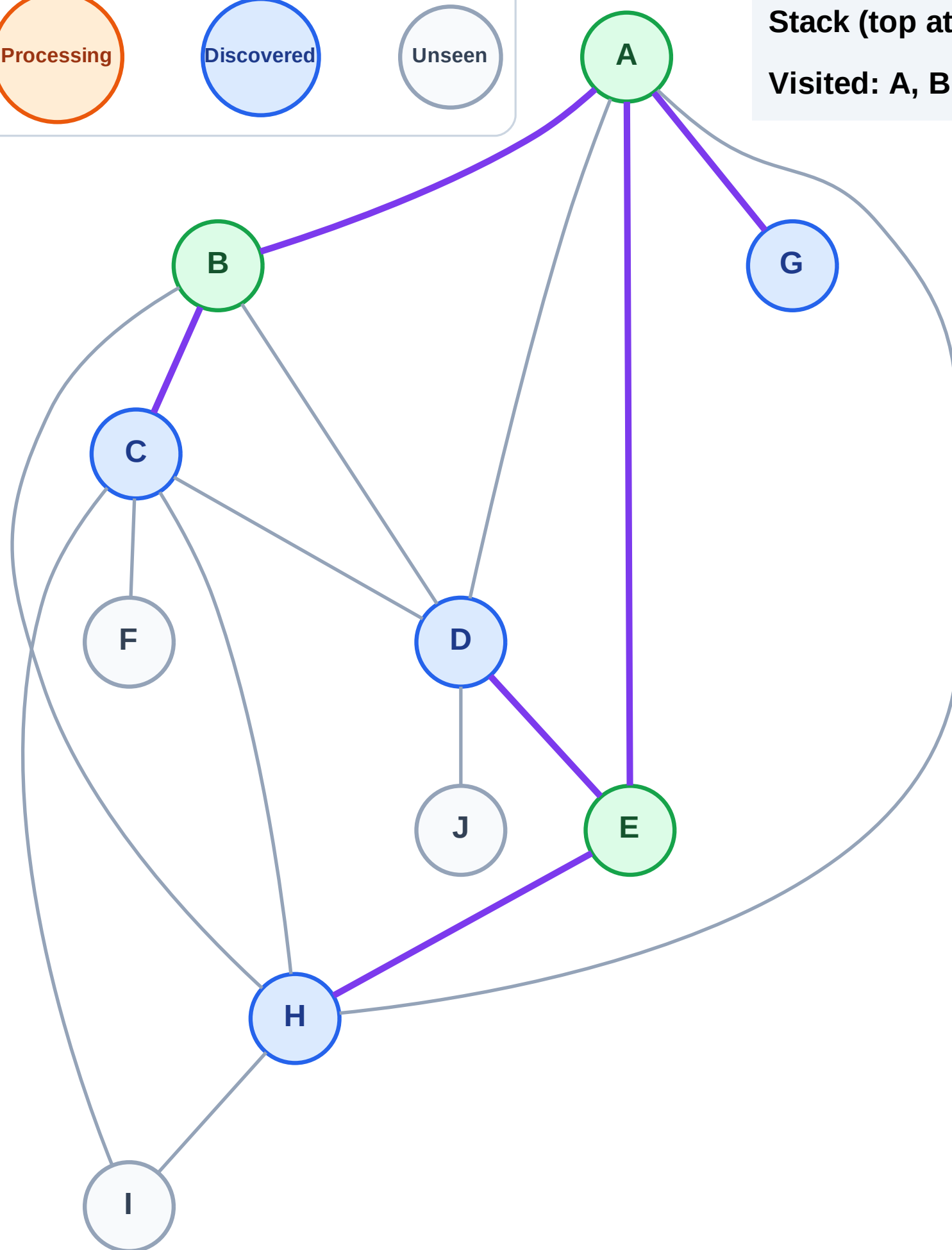
DFS Traversal

Step 7: Discover C from B



Stack (top at right): H | D | G | C

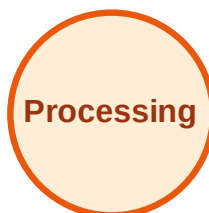
Visited: A, B, E



DFS Traversal

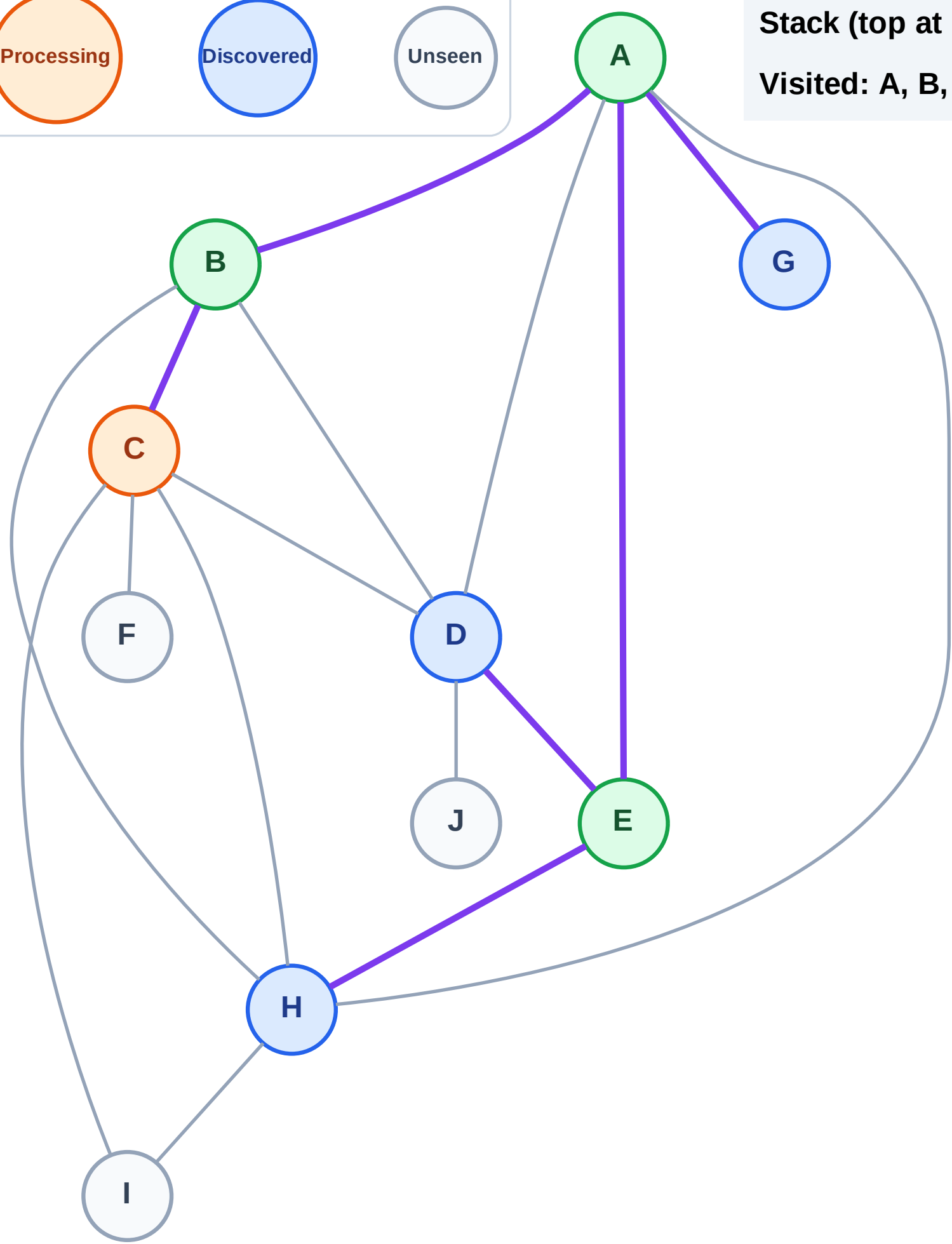
Step 8: Process node C

Legend



Stack (top at right): H | D | G

Visited: A, B, E



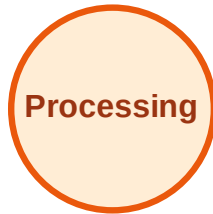
DFS Traversal

Step 9: Discover I, F from C

Legend



Complete



Processing



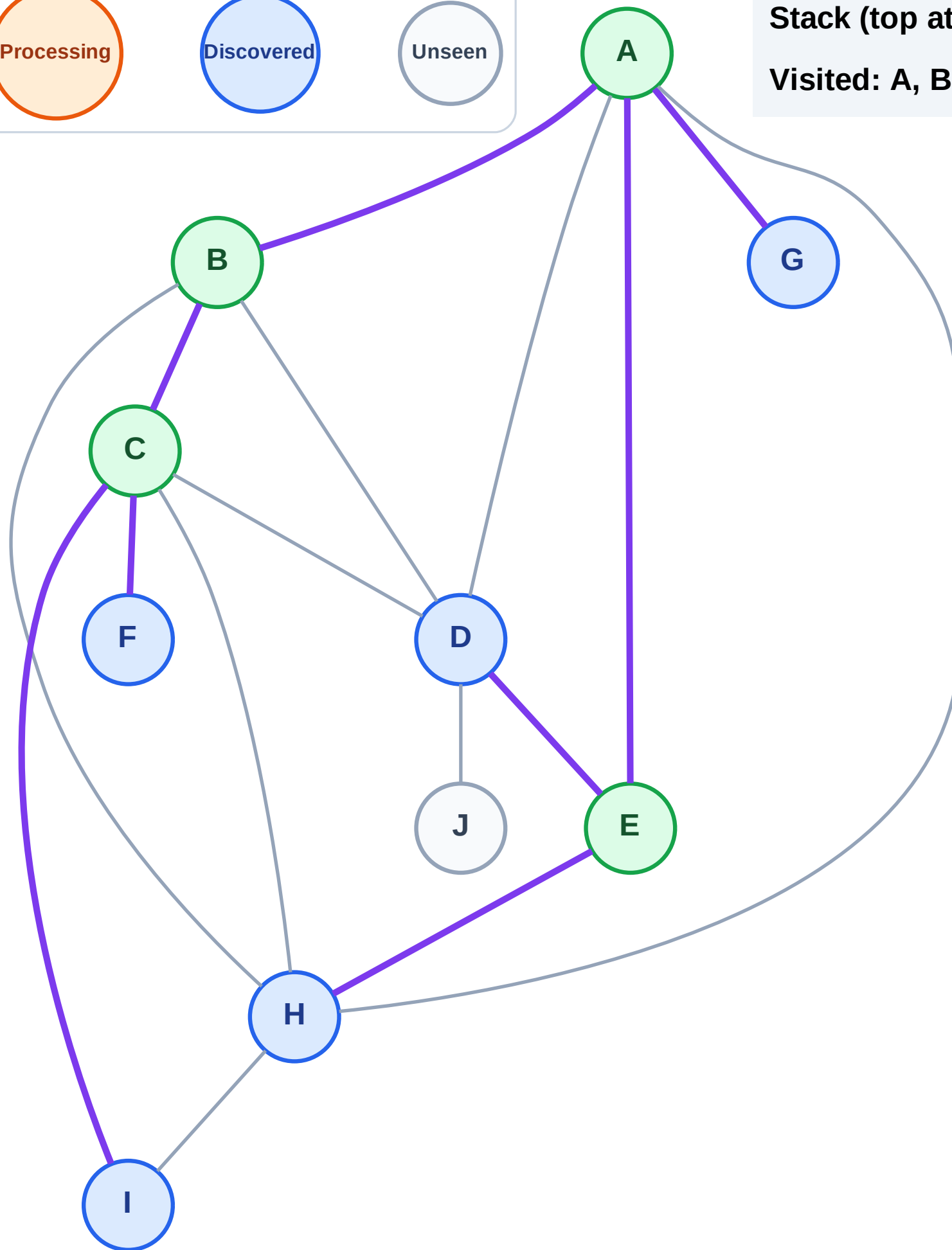
Discovered



Unseen

Stack (top at right): H | D | G | I | F

Visited: A, B, C, E



DFS Traversal

Step 10: Process node F

Legend



Complete



Processing



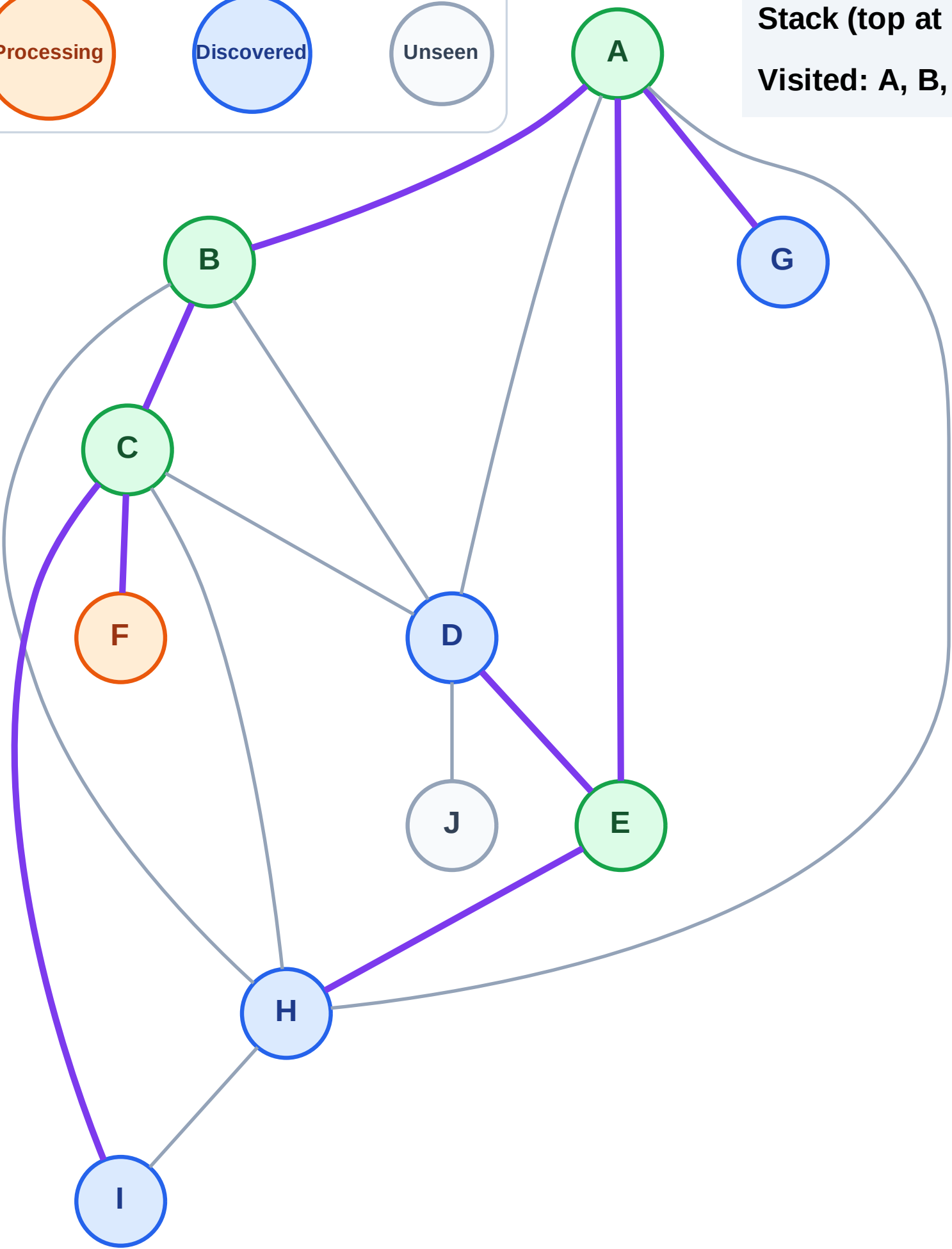
Discovered



Unseen

Stack (top at right): H | D | G | I

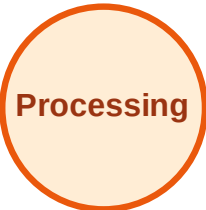
Visited: A, B, C, E



DFS Traversal

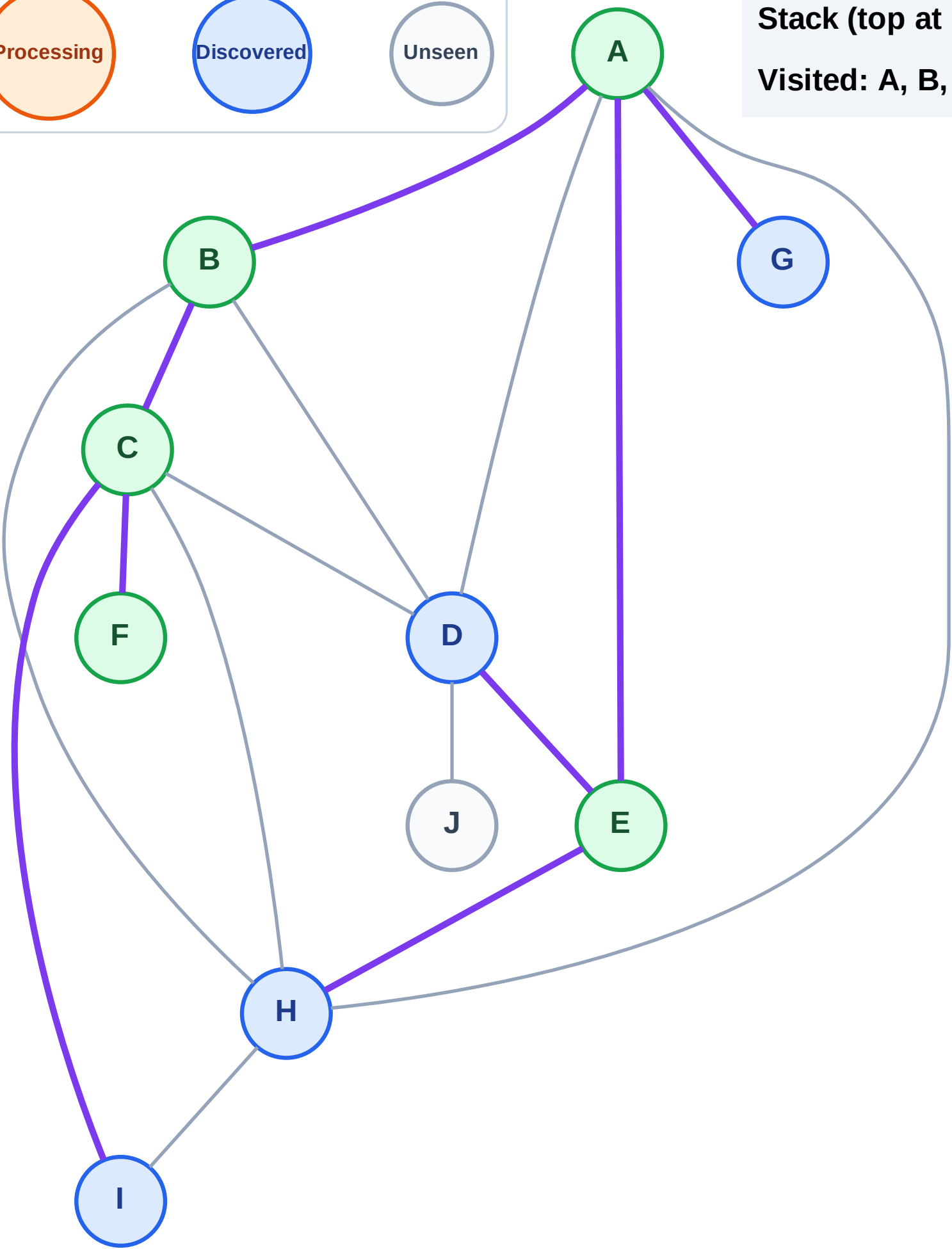
Step 11: No new neighbors from F

Legend



Stack (top at right): H | D | G | I

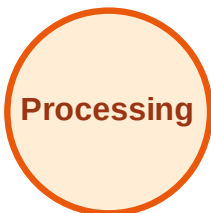
Visited: A, B, C, E, F



DFS Traversal

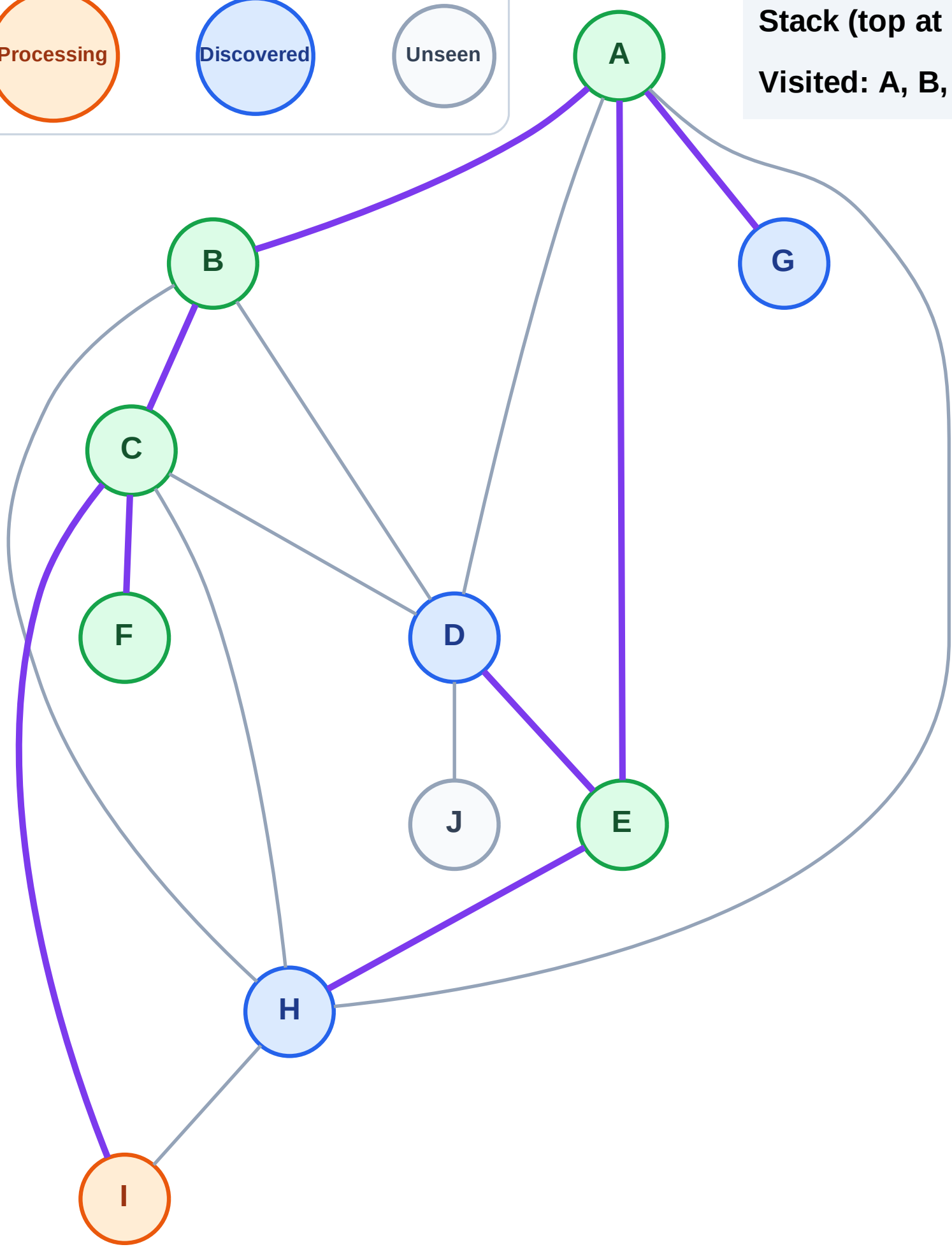
Step 12: Process node I

Legend



Stack (top at right): H | D | G

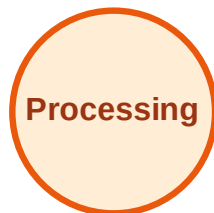
Visited: A, B, C, E, F



DFS Traversal

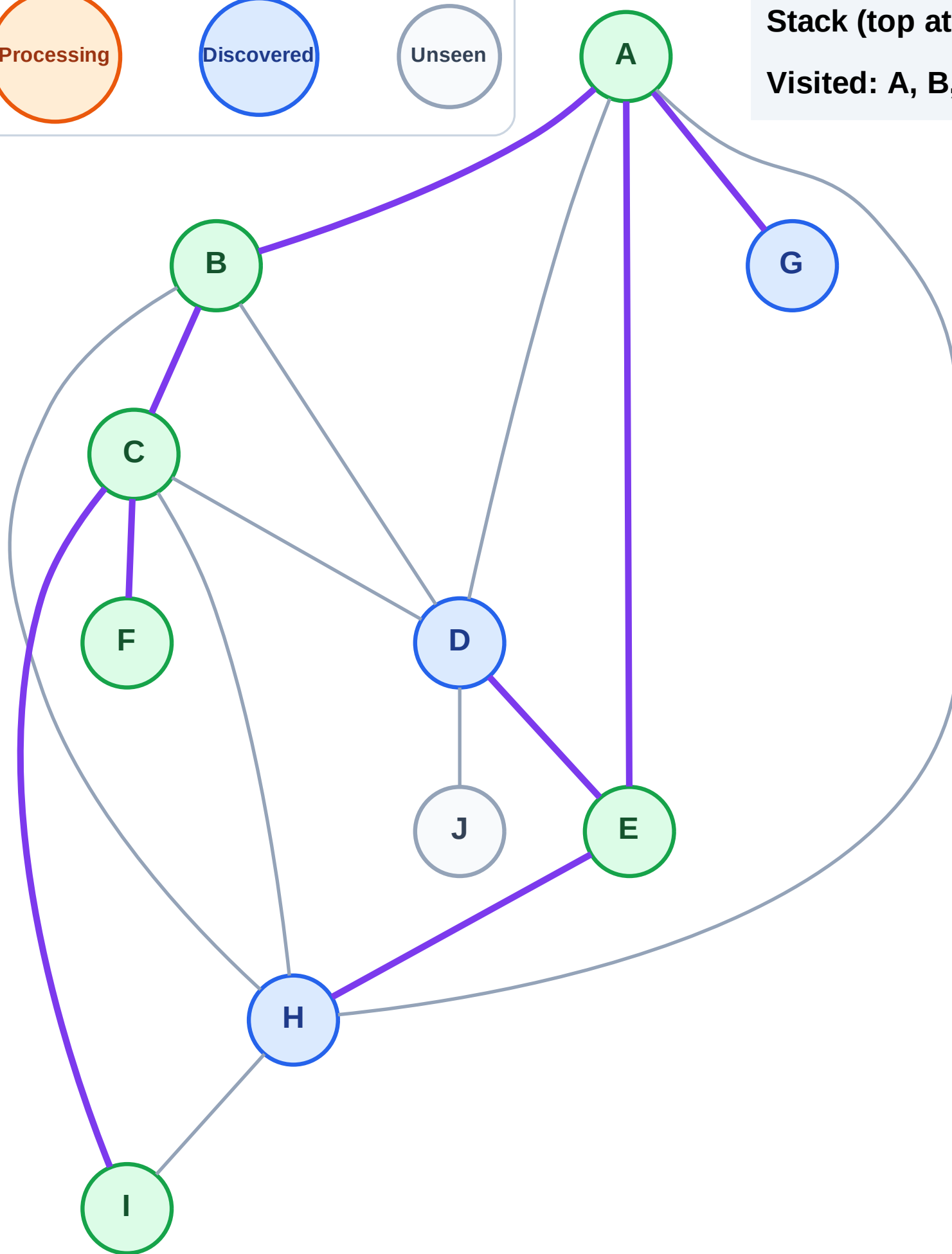
Step 13: No new neighbors from I

Legend



Stack (top at right): H | D | G

Visited: A, B, C, E, F, I



DFS Traversal

Step 14: Process node G

Legend

Complete

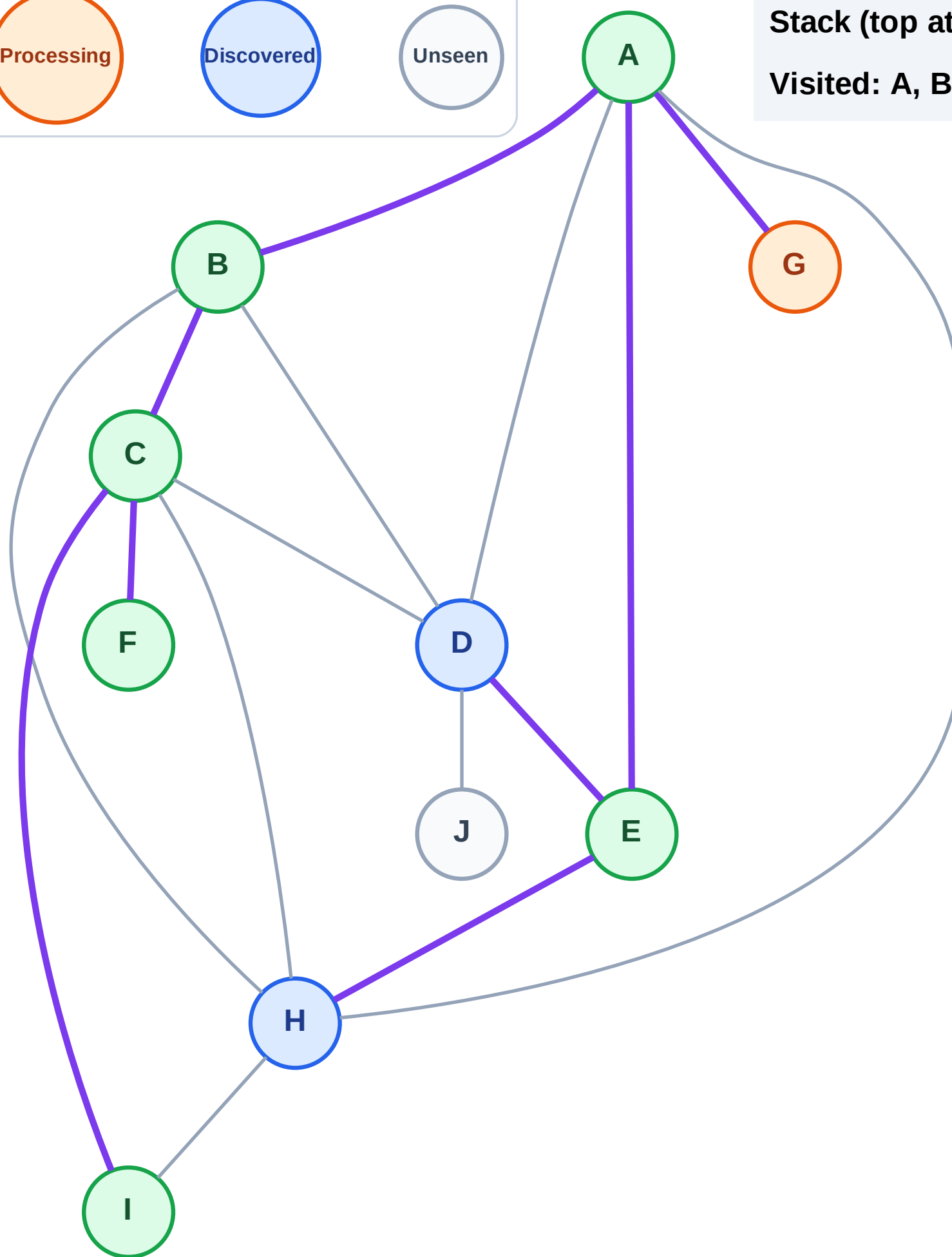
Processing

Discovered

Unseen

Stack (top at right): H | D

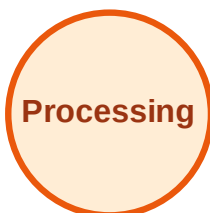
Visited: A, B, C, E, F, I



DFS Traversal

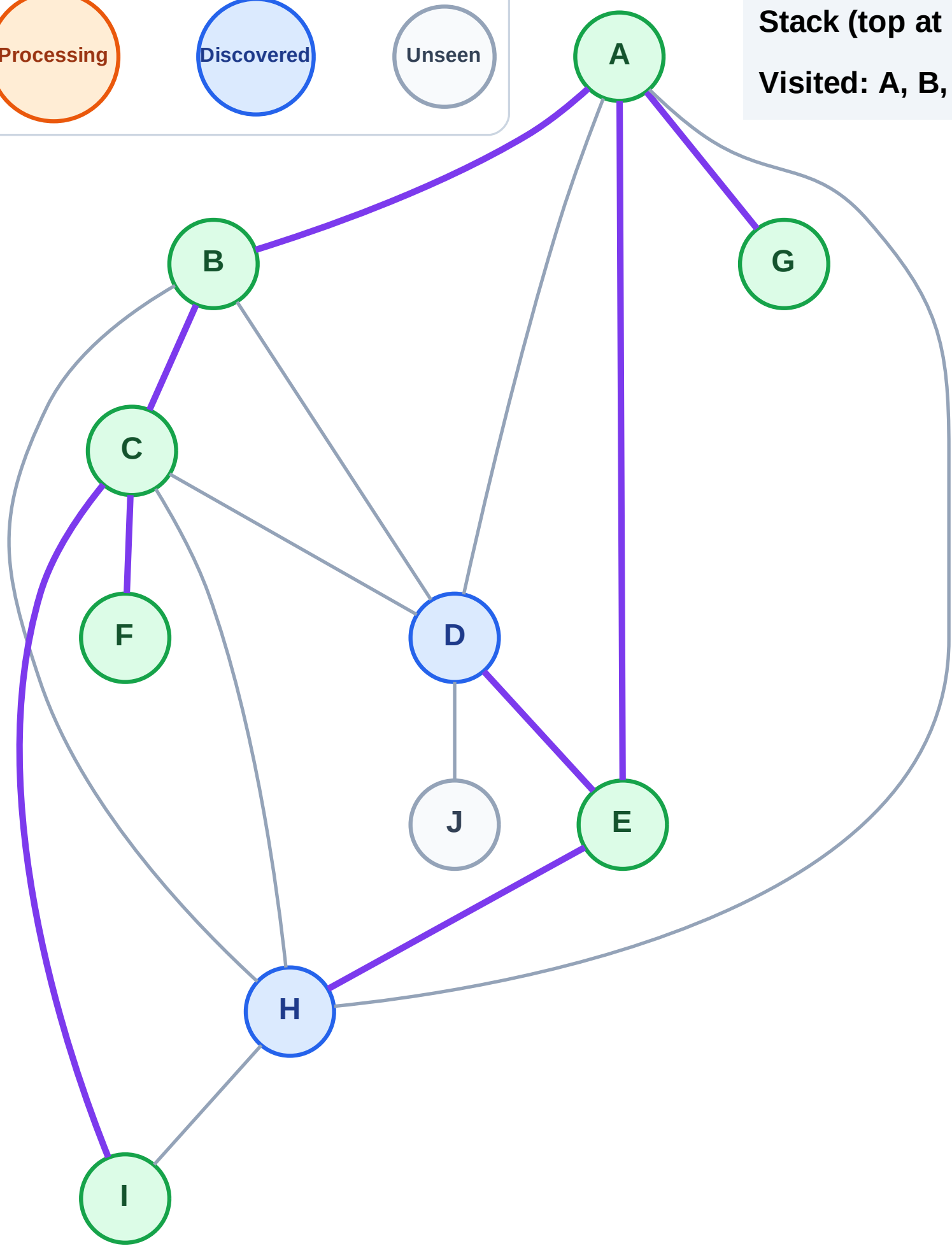
Step 15: No new neighbors from G

Legend



Stack (top at right): H | D

Visited: A, B, C, E, F, G, I



DFS Traversal

Step 16: Process node D

Legend

Complete

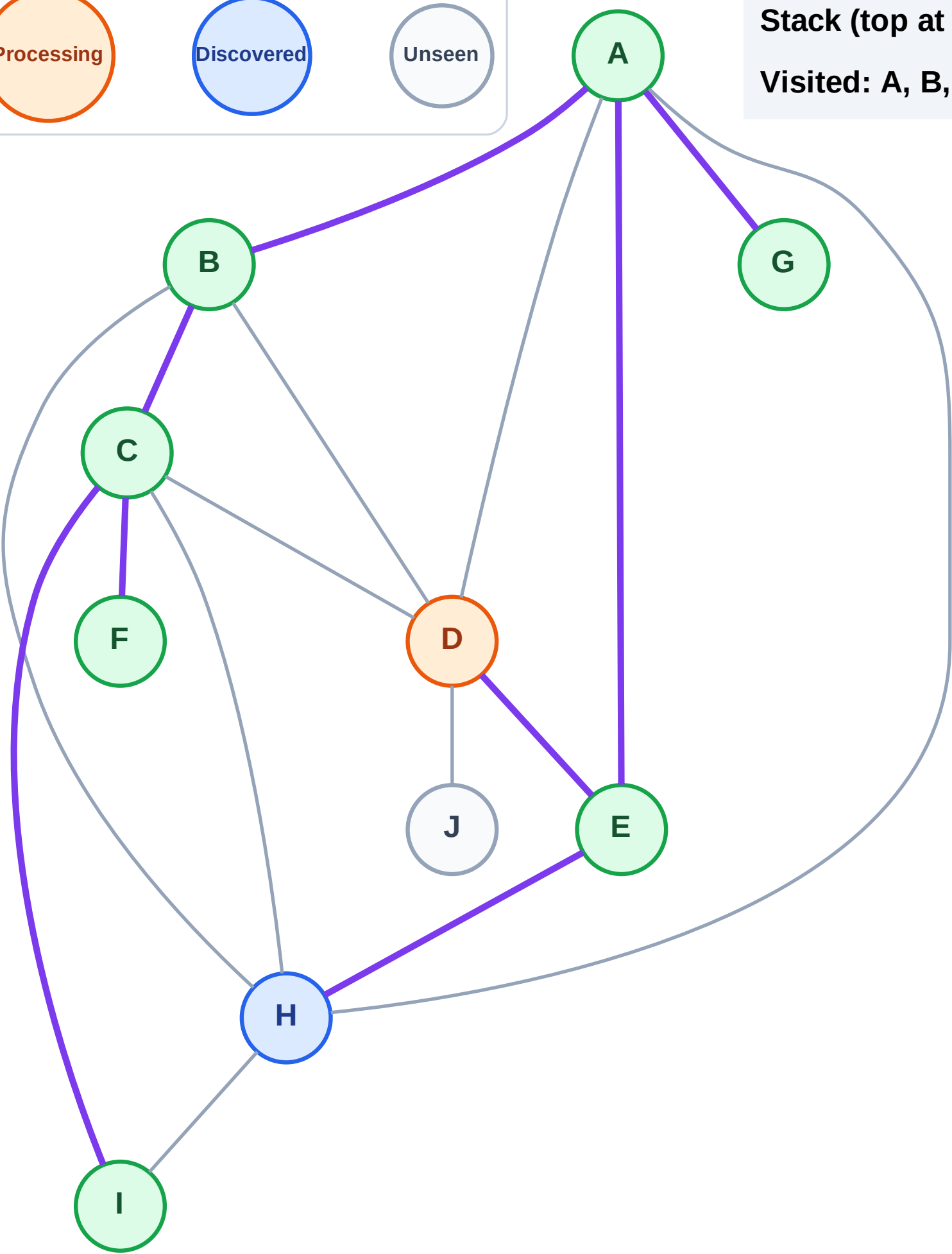
Processing

Discovered

Unseen

Stack (top at right): H

Visited: A, B, C, E, F, G, I



DFS Traversal

Step 17: Discover J from D

Legend

Complete

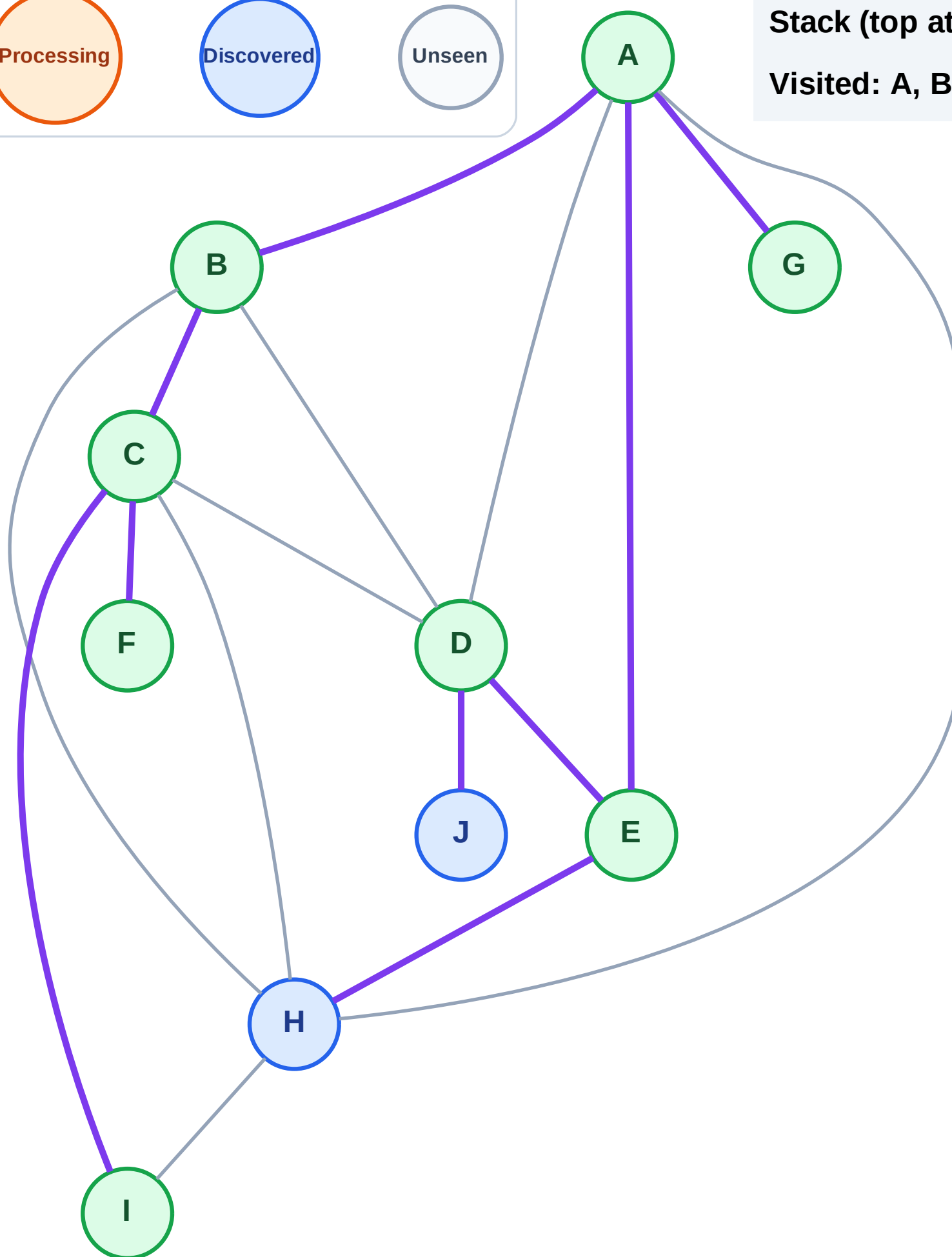
Processing

Discovered

Unseen

Stack (top at right): H | J

Visited: A, B, C, D, E, F, G, I



DFS Traversal

Step 18: Process node J

Legend

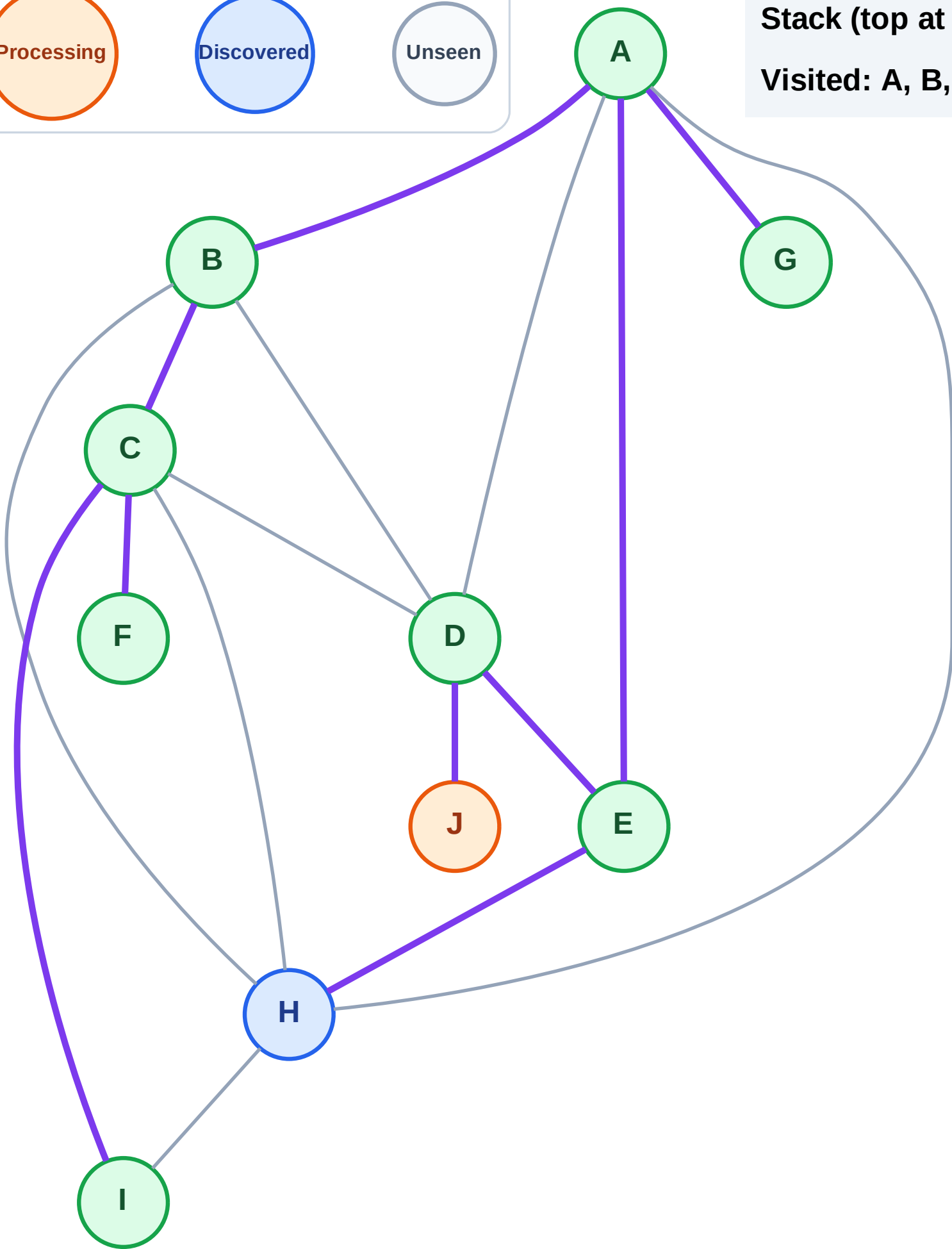
Complete

Processing

Discovered

Unseen

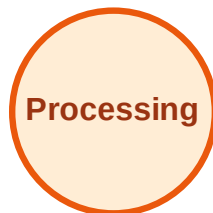
Stack (top at right): H
Visited: A, B, C, D, E, F, G, I



DFS Traversal

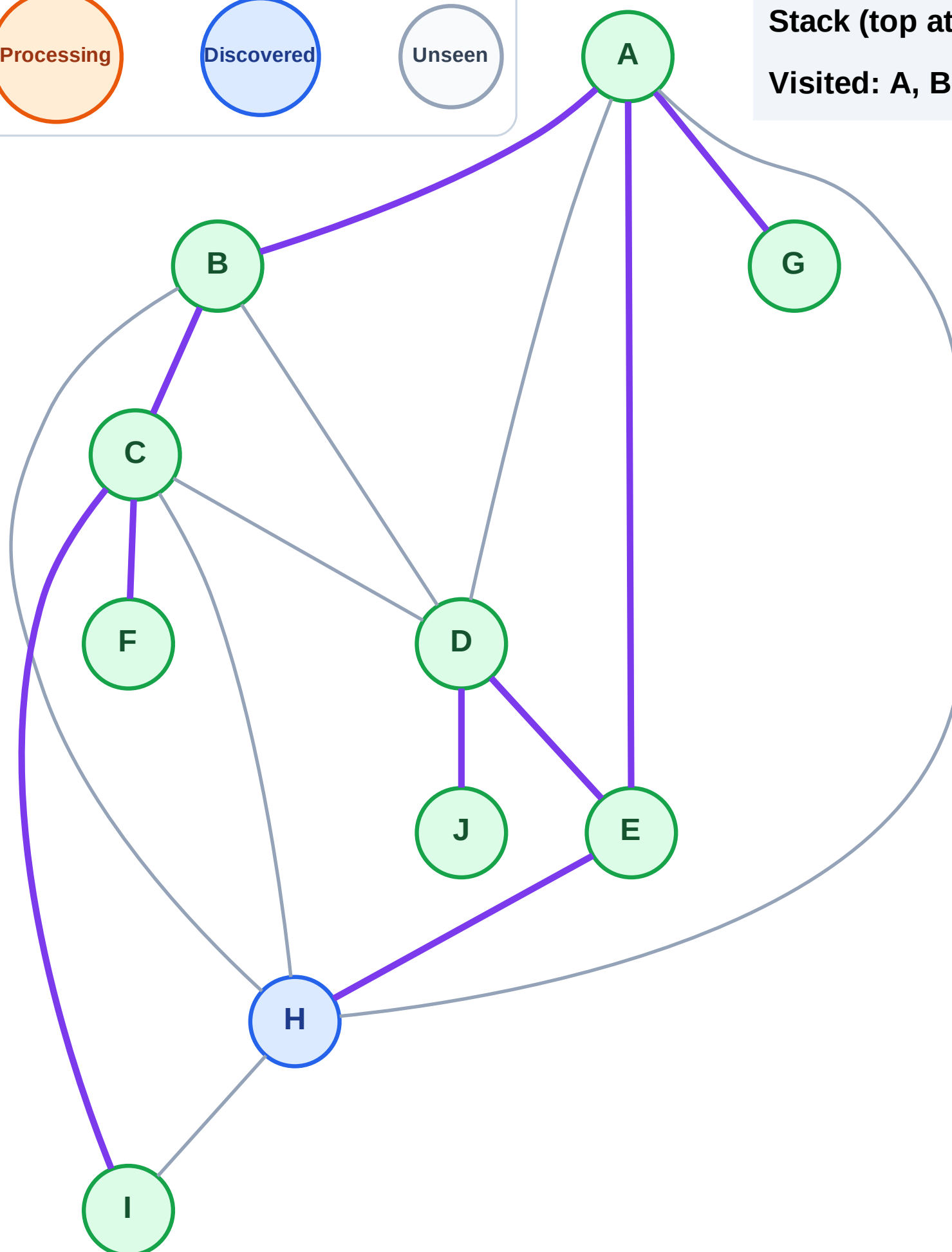
Step 19: No new neighbors from J

Legend



Stack (top at right): H

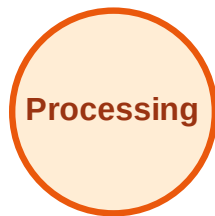
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

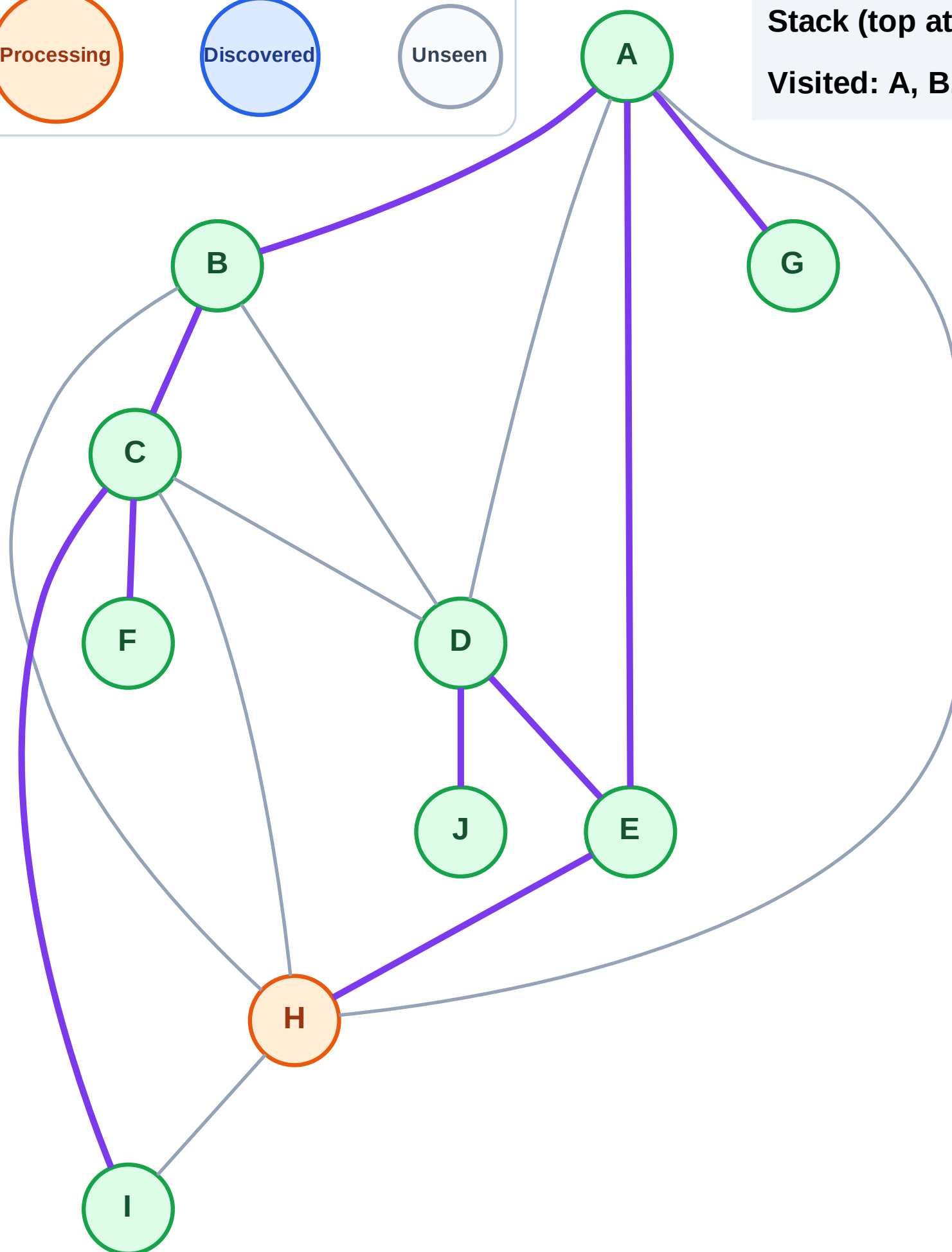
Step 20: Process node H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, I, J



Step 21: No new neighbors from H

Step 21: No new neighbors from H

